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United States Patent	12398389
Kind Code	B2
Date of Patent	August 26, 2025
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Methods for isolating nucleic acids with size selection

Abstract

Disclosed here is a method for isolating nucleic acids from a biological sample, comprising: (a) contacting a first composition comprising nucleic acids obtained from a biological sample with a first matrix under a low-stringency binding condition having less than 1% aliphatic alcohols that binds less than 5% of nucleic acids of shorter than about 118 bp and more than 30% of nucleic acids longer than about 194 bp to a first matrix; and (b) contacting a second composition comprising remainder of the first composition with a second matrix under a high-stringency binding condition having less than 1% aliphatic alcohol that binds more than 70% of nucleic acids longer than about 72 bp and 30% of nucleic acids longer than about 50 bp to the second matrix. Further disclosed is a kit for isolating nucleic acids from a biological sample, comprising (a) a first binding buffer for establishing a low-stringency binding condition having less than 1% aliphatic alcohols that binds less than 5% of nucleic acids shorter than about 118 bp and more than 30% of nucleic acids longer than about 194 bp to a matrix, and (b) a second binding buffer for establishing a high-stringency binding condition having less than 1% aliphatic alcohol that binds more than 70% of nucleic acids longer than about 72 bp and 30% of nucleic acids longer than about 50 bp to the matrix.

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Appl. No.:	16/969892
Filed (or PCT Filed):	February 15, 2019
PCT No.:	PCT/US2019/018274
PCT Pub. No.:	WO2019/161244
PCT Pub. Date:	August 22, 2019

Prior Publication Data

Document Identifier

US 20210009990 A1

Publication Date

Jan. 14, 2021

Related U.S. Application Data

continuation-in-part parent-doc WO PCT/US2018/018425 20180215 PENDING child-doc US 16969892

us-provisional-application US 62631336 20180215

Publication Classification

Int. Cl.: C12N15/10 (20060101); C12Q1/6853 (20180101)

U.S. Cl.:

CPC C12N15/1013 (20130101); C12Q1/6853 (20130101);

Field of Classification Search

CPC: C12N (15/1013); C12N (15/1017); C12N (15/1006); C12Q (1/6853); C12Q (2527/125)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage Entry of PCT/US19/18274, filed Feb. 15, 2019, which claims the benefit of U.S. Provisional Application Ser. No. 62/631,336, filed Feb. 15, 2018, and PCT Application Ser. No. PCT/US2018/18425, filed Feb. 15, 2018. PCT Application Serial No. PCT/US2018/18425 claims the benefit of U.S. Provisional Application Ser. No. 62/461,735, filed Feb. 21, 2017. Each of these applications cited above is hereby incorporated by reference in its entirety.

BACKGROUND

(1) Non-invasive and minimally invasive liquid biopsy tests utilize sample material collected from external secretions or by needle aspiration for analysis. The extracellular nuclear DNA present in the cell-free fraction of bodily fluids such as urine, saliva and other glandular secretions, cerebrospinal and peritoneal fluid, and plasma or serum from blood, contain sufficient amounts of target sequences to support accurate detection of genetic anomalies that underlie many disorders that could otherwise be difficult or impossible to diagnosis outside of expensive medical biopsy procedures bearing substantial risk. In blood, the circulating cell free DNA (cfDNA) fraction represents a sampling of nucleic acid sequences shed into the blood from numerous sources which are deposited there as part of the normal physiological condition. The origin of a majority of cfDNA can be traced to either hematological processes or steady-state turnover of other tissues such as skin, muscle, and major organ systems. Of great clinical importance was the discovery that a significant and detectable fraction of cfDNA derives from exchange of fetal DNA crossing the placental boundary and from immune-mediated, apoptotic or necrotic cell lysis of tumor cells or cells infected by viruses, bacterium, or intracellular parasites. This makes plasma an extremely attractive specimen for molecular analytical tests and in particular, test that leverage the power of deep sequencing for diagnosis and detection.

(2) Physical separation by molecular sieving has been exploited many times and in many ways to characterize DNA. The most common sieving techniques that separate DNA by size are electrophoretic in nature. These include native fragment analysis in agarose gels or capillary electrophoresis (CE) through polymer supports, or denaturing methods in polyacrylamide-urea gels or through urea polymer CE, so extensively used in Maxim-Gilbert and Sanger sequencing. Chromatographic separation of DNA using ion exchange (IE) or reverse phase (RP) supports is also widely used to characterize or purify DNA. IE and RP methods are routinely used to separate conjugates from non-conjugates and unincorporated label following covalent modification with for example, reactive amines, sulfhydryl or azido groups, and ligands such as biotin or fluorescent dyes.

These techniques depend on the chemical differences imparted by the presence of the particular substituent, which typically alter charge and/or hydrophobicity of the DNA-adduct relative to unlabeled DNA. Molecular sieving and chromatographic techniques rely on physical-hydrodynamic differences associated with DNA length, or chemical-physical difference associated with covalent modification.

(3) There are other techniques that achieve size dependent fractionation of native DNA based on size, that do not depend on sieving or chemical differences, but which operate by differential adsorption to solid supports. The most famous and highly applied approach is Solid Phase Reversible Isolation (SPRI) selection which utilizes carboxyl coated paramagnetic beads in the presence of high salt and the crowding agent polyethylene glycol (PEG), to promote controlled adsorption, tuned for a given size by the varying PEG concentrations. DNA molecules of differing length can be partitioned by subjecting source DNA to various binding and elution schemes in the presence of different amounts of PEG. This size selection method is routinely applied to separate PCR primers or un-ligated adapters smaller in size than PCR amplified or ligation products. It has also been used to fractionate sheared genomic DNA (gDNA) and even “clean up” purified cell-free DNA (cfDNA) by removing larger contaminating gDNA prior to molecular analysis or sequencing library preparation. In all cases, the input for SPRI-based selection fit one or more of the following criterion: (1) the input DNA has already been purified from the biological sample; (2) the volume is relatively small (e.g., 50 to 100 μ L); and/or (3) the DNA exists in a defined composition (e.g., highly pure in buffer, or in reaction conditions such as end repair, ligation, PCR amplification, etc.).

SUMMARY

(4) Disclosed here is a purification size selection method that is performed concomitant with nucleic acid isolation directly from a complex biological sample as the first step in sample preparation for molecular analysis. The does not require the input of previously purified DNA to achieve fine discrimination in the small DNA size regime (50 to 300 bp), and can be adjusted for large volume (10 to 20 mL) of liquid sample (e.g., human plasma, serum, or urine). The method is based on selective adsorption to a solid support. The workflow described herein encompasses simultaneous purification and size selection of cfDNA by sequential low and high stringency binding and elution, and is comprised of the following elements; (1) proteolysis and establishment of a low stringency binding condition; (2) DNA immobilization, washing, and first elution; (3) establishment of a high stringency binding condition; and (4) DNA immobilization, washing, and second elution. The size distribution of cfDNA preserved in the first and second elution reflects the differential binding affinity of long versus short dsDNA fragments that is dependent on the underlying low and high stringency binding conditions established. The net result is a partitioning of longer cfDNA fragments to the first eluate and shorter cfDNA to the second eluate. Both are preserved in the process and due to the extremely high capture efficiency, little loss of the starting DNA from the sample is lost, and therefore all molecules can be processed for analysis post purification/selection.

(5) Accordingly, in one aspect, the inventions described herein relate to a method for isolating nucleic acids from a biological sample, comprising: (a) contacting a first composition comprising nucleic acids obtained from a biological sample with a first matrix under a low-stringency binding condition that comprises less than 1% aliphatic alcohols, that binds less than 5% of nucleic acids of about 72 bp or shorter and more than 30% of nucleic acids of about 194 bp or longer to the first matrix; and (b) contacting a second composition comprising remainder of the first composition with a second matrix under a high-stringency binding condition that also comprises less than 1% aliphatic alcohols, that binds more than 70% of nucleic acids of about 72 bp or longer and more than 30% of nucleic acids of about 50 bp or longer to the second matrix.

(6) In another aspect, the inventions described herein relate to a kit for isolating nucleic acids from a biological sample, comprising (a) a first binding buffer for establishing a low-stringency binding condition that comprises less than 1% aliphatic alcohols, that binds less than 5% of nucleic acids of

about 72 bp or shorter and more than 30% of nucleic acids of about 194 bp or longer to a matrix, and (b) a second binding buffer for establishing a high-stringency binding condition that also comprises less than 1% aliphatic alcohols, that binds more than 70% of nucleic acids of about 72 bp or longer and more than 30% of nucleic acids of about 50 bp or longer to the matrix. (7) These and other features, together with the organization and manner of operation thereof, will become apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows recovery of DNA as a function of the concentration of isopropanol (IPA) and guanidine chloride (GnCl) in the nucleic acid binding state. Recovery of short dsDNA increased as the IPA concentration was raised and GnCl concentrations fell as a certain amount of volume was displaced by the added solvent. Exogenous DNA targets were spiked after plasma proteolysis and quantified by real time PCR and standard curve methods. Percent recovery of target fragments was determined by comparing against spike controls assembled by adding the original spike amount to eluates recovered from matched plasma samples by DNA isolation methods similar to the test method. Each test sample was normalized with buffer to account for the volume of spike targets added to recovery controls.

(2) FIG. 2 shows recovery of DNA as a function of the concentration of acetonitrile (ACN) and GnCl in the nucleic acid binding state. Recovery of short dsDNA increases as the ACN concentration is raised and GnCl concentrations fall due to volume displaced by the added solvent. Exogenous DNA targets spiked following proteolysis were quantified by real time PCR quantified using standard curve methods. The percent recovery for each target was determined by comparison against recovery controls assembled by adding the original spike amount to eluates recovered from matched plasma samples isolated with a similar test chemistry. Test samples were normalized with buffer to account for spike volumes added to recovery controls.

(3) FIG. 3 shows recovery of DNA as a function of GnCl concentration in the nucleic acid binding state when ACN was held constant at 8.3%. Recovery of short dsDNA increased as the concentration of GnCl was increased relative to a constant amount of ACN. Exogenous DNA targets were spiked following proteolysis and quantified by real time PCR and standard curve methods. The percent recovery of each target fragment was calculated by comparing against spike controls in which the original spike amount was added to eluates recovered from matched plasma samples. All test samples were normalized with buffer to account for spike volumes added to recovery controls.

(4) FIG. 4 shows recovery of DNA as a function of the relative amount of ACN and GnCl in the nucleic acid binding state. Recovery of short dsDNA decreased as the concentration of GnCl or percentage of ACN was reduced in the nucleic acid binding state. An increase of either ACN or GnCl compensated for the deficiency of the other. For instance, recovery of the 72 bp fragment improved when ACN was held constant at 11% and GnCl increased from 4.19 M to 4.47 M. This also happened when GnCl was held constant at 4.19 M and ACN increased from 11% to 16.6%. Exogenous DNA targets were spiked following proteolysis and quantified by real time PCR using standard curve methods. Percent recovery for each fragment was determined by comparison to spike controls established by adding the original spike amount to eluted cfDNA isolated from plasma samples by a similar test method. All test samples were normalized with buffer to account for the volume added to recovery controls.

(5) FIG. 5 shows recovery of DNA with increasing amount of Triton X100 in the nucleic acid binding state. Increased concentration of Triton yielded an increase in spike recovery of the 72 bp

exogenous DNA target. Targets were quantified by real time PCR using standard curve methods. Percent recovery was determined by comparison against spike controls generated by adding spike targets to eluted cfDNA isolated from plasma samples by a similar test method. All test samples were normalized with buffer to account for the addition of spike material to recovery controls.

(6) FIG. 6 shows recovery of DNA from 2 extractions from 2 single donor plasma samples (N=4). The amount of Triton X100 in the nucleic acid binding state was varied to reveal an overall increase in spike recovery with increased Triton at two different ACN and GnCl concentrations in the nucleic acid binding state. Exogenous DNA targets spiked following proteolysis were quantified by real time PCR using standard curve methods. Percent recovery for each target fragment was determined by comparison against the original amount of spike target added to recovered plasma cfDNA extracted by a similar test method. All test samples were normalized with buffer to account for the addition of spike material to recovery controls.

(7) FIG. 7 shows recovery of DNA as a function of organic solvent used to establish the nucleic acid binding state. Various water soluble solvents were added to create the binding condition, and surprisingly it was revealed that solvents whose results are presented (ACN, PCN, BCN, IBCN and THF) all promoted an increase in recovery of the 72 bp and 118 bp fragments compared to the 0% controls (series 2, 4 and 6) in which solvent was replaced by water. Exogenous DNA targets spiked following proteolysis were quantified by real time PCR using standard curve methods. Percent recovery for each target fragment was determined by comparison against the original amount of spike target added to recovered plasma cfDNA extracted by a similar test method. All test samples were normalized with buffer to account for the addition of spike material to recovery controls.

(8) FIG. 8 shows a generalized Plasma ccfDNA Extraction Workflow.

(9) FIG. 9 Details the increased recovery of fetal cfDNA by when ACN and GnCl are present in the nucleic acid binding state, as revealed by NIPT analysis. Fetal fraction estimates derived from the ratio of fetal to maternal SNPs are shown. The pairwise comparison included 16 maternal plasma samples isolated with two different optimized methods, one utilizing acetonitrile (ACN) and one isopropyl alcohol (IPA), to establish the nucleic acid binding state. Differences in fetal fraction ((ACN)-(IPA)) are shown above each matched pair. A paired t-test reveals a statistically significant increase ($t=0.003$) when acetonitrile was used to establish the nucleic acid binding state.

(10) FIG. 10 presents a one-way analysis of variance showing decreased filtration times for plasma cfDNA extractions for which the binding state was established with acetonitrile (ACN) and compared directly to isopropyl alcohol (IPA). The pairwise comparison is of 16 plasma samples from maternal donors isolated by two different optimized methods, one utilizing acetonitrile (ACN) and one isopropyl alcohol (IPA) to establish the nucleic acid binding state (Refer to FIGS. 1 and 2 for a comparison of yield of fragments of various size). Mean filtration times were much shorter when the aprotic solvent ACN was used.

(11) FIG. 11 shows extraction linearity. The plot summarized the total yield of cfDNA and % recovery of the 72, 118, 194 and 1078 bp spike targets from varying input amounts of human plasma. Human plasma; 1, 2, 5, 10, 15 and 20 mL were used as input. The 1 to 5 mL plasma samples were normalized to 10 mL with the addition of 1×PBS and extracted, along with the 10 mL plasma sample, by the standard 10 mL plasma NAS protocol. Reagent volumes were increased proportionally for the 15 and 20 mL plasma samples. 200 pg of spike target mixture was added to each normalized plasma and recovery, as a % of control, was determined by qPCR. Recovery of cfDNA was estimated from Caliper LabChip CE traces by quantifying DNA between 120 and 220 bp (i.e., mono-nucleosome in size). The results show that DNA extraction efficiency is consistent across all plasma volumes. This is shown in the upper portion of the plot by the clustering of recovery data for all four DNA fragment, which returned 80 to 92% of the original spike amount (right axis), for the 72 bp (solid line), 118 bp (dot-dash-dot), 194 bp (dotted line), and 1078 bp (dashed line), respectively. Recovery of monosomal cfDNA scaled linearly with plasma volume input, and correspondingly, the recovery per mL of plasma was constant from 1 to 20 mL of

plasma, demonstrating that this method is scalable and efficient across a broad range of input.

(12) FIG. 12 shows extraction recovery of 25 bp, 50 bp and 75 bp dsDNA fragments from plasma. A mixture of these dsDNA spike fragments was added to a 10 mL Test plasma sample, and buffer only to a matched 10 mL plasma to serve as Control. The NAS extraction method was carried out on both samples. An equivalent amount of spike fragments were added eluted cfDNA from the Control and the same amount of buffer to the Test eluate. 1 uL of each eluate was separated by capillary electrophoresis on an Agilent Bioanalyzer HS chip and the fluorescence plots for each run were overlayed. The plot of the Control run shows three of the tallest peaks at 25, 50 and 75 bp. The plot of lower amplitude shows that 80.5% and 44.2% of the 75 and 50 bp fragments, respectively, were recovered from 10 mL plasma by the NAS extraction method. None of the 25 bp fragment, however, was recovered. This suggests that the NAS chemistry is capable of recovering fragments as small as 50 bp with reasonable efficiency, and fragments of 75 bp in length with good efficiency, in agreement with the qPCR results shown in other figures.

(13) FIG. 13 shows a comparative workflow of a standard, single or one pass, cfDNA extraction process. The method shown establishes a high-stringency binding condition to isolate and purify DNA from a biological sample.

(14) FIG. 14 shows an exemplary workflow of dual filtration size selection using low and high stringency binding buffers (LSBB, HSBB). The method shown is a two-step process wherein the pass through fraction of the 1.sup.st, low stringency, matrix contacting step serves as the input for the 2.sup.nd, high stringency, matrix contacting step. Following washes to remove contaminants, the purified nucleic acids bound to both 1.sup.st and 2.sup.nd pass contact matrices can be eluted.

(15) FIG. 15 shows data from a proof-of-concept experiment in which the second passage of acidified lysate was able to recover small fragments missed in the first passage.

(16) FIG. 16 is an electropherogram showing plasma cfDNA and co-purified defined fragments. The 72 bp, 118 bp, 194 bp, and 1078 bp fragments tracked by qPCR are shown. The main cfDNA peak falls between 118 and 310 bp, 194 bp fragment falls within the leading edge. Fragments from di- and tri-nucleosomes are bigger.

(17) FIG. 17 shows data from an experiment that tested low stringency binding conditions by uniform titration of binding buffer to progressively increase the stringency. The proteolysis conditions (PKDB+PK) are kept the same.

(18) FIG. 18 shows data from an experiment that tested increasing the pH of the low stringency binding condition. The pH 9 adjusted binding buffer (BB) promoted improved matrix retention of fragments 194 and 1078 bp in length, but did not retain fragments 72 and 118 bp in length compared to: compare 1.sup.st pass recovery of the 118 bp fragment of rows 1 and 2 with that of rows 7 and 8, where both received 5 mL of BB at the normal pH and at pH 9, respectively.

(19) FIG. 19 shows (top) data from an experiment that tested low stringency binding-high stringency binding system with ACN or without ACN Rows 3 and 5 show the improved 2.sup.nd pass recovery of the 72, 118 and 194 bp fragments in the presence of ACN. The bottom table shows data from an experiment that tested further increasing the pH of the low stringency binding-high stringency binding system that suggest better 1.sup.st pass retention of the larger 1078 bp fragment with pH 9 binding buffer (row 3 vs row 5).

(20) FIG. 20 shows data from an experiment applying the two-step filtration method to preferentially enrich for fetal cfDNA from 12 plasma samples from pregnant mothers. The right panel presents the experimental scheme by which maternal plasma was divided equally and extracted by the standard one pass method in parallel with the two-step method. cfDNA from the one pass, and 1.sup.st and 2.sup.nd pass of the 2-step method, served as input for library construction preceding SNP based, next generation sequencing based, prenatal genotypic analysis. Essentially all 1078 bp fragments were captured in the first Pass; nearly all 72 bp fragments were captured in second pass eluate; almost no 194 bp and 1078 bp fragments were captured in second pass eluate; and variable percentage of the 118 bp fragment were captured in second pass eluate.

cfDNA is presumed to follow the same pattern of size discrimination.

(21) FIG. 21 shows graphically the percentage recovery data of FIG. 20.

(22) FIG. 22 tabulates child fraction estimate (CFE) from the experiment shown in FIG. 20, applying the two-step filtration method to preferentially enrich for fetal cfDNA from 12 plasma samples from pregnant mothers. Nearly all 72 bp fragments are captured in second pass eluate, while almost no 194 bp and 1078 bp fragments are captured in second pass eluate. % CFE increased in second pass eluate.

(23) FIG. 23 shows graphically the CFE data of FIG. 22. Nearly all 72 bp fragments are captured in second pass eluate; almost no 194 bp and 1078 bp fragments are captured in second pass eluate; and variable 118 bp fragment are captured in second pass eluate. Absolute % CFE increase ranged between 0.8% to 11.8%, with a mean increase of 4.19% and median increase of 2.51%.

(24) FIG. 24 shows heterate plots for maternal plasma sample #3, which clearly shows % CFE increase due to second pass selected cfDNA. The child SNPs are seen clearly in the fetal cfDNA enriched plot (middle, 2.sup.nd pass) compared to the control plot (top, one pass method) and depleted plot (bottom, 1.sup.st pass).

(25) FIG. 25 is an electropherogram overlay showing library traces produced from maternal plasma sample #1's control, first pass, and second pass. Library profiles reflect the underlying cfDNA populations in selected fractions. There was a detectable 6 bp of separation between the center peaks of 1.sup.st and 2.sup.nd pass libraries and each bracketed the unselected "control" library peak. The shoulders at the leading edge (2.sup.nd Pass) and trailing edge (1.sup.st pass) show an even greater degree of separation; ~230 bp compared to ~270 bp.

(26) FIG. 26 shows graphically the No Salt Increase-Size Selection At Purification (NSI-SSAP) workflow employing dual filtration size selection under low and high stringency binding conditions in which salt concentration was not increased. The concentration of chaotropic salt in the low stringency binding condition begins at >30% and remains unchanged, or decreases slightly, when the subsequent high stringency binding condition is formed. To begin, an initial lysis portion is established by combining 10 mL plasma from whole blood with 200 μ L of 20 mg/mL Proteinase K and 11.7 mL of low stringency binding buffer (LSBB) in a 50 mL conical tube. The mixture is incubated at 42° C. for 45 minutes to digest plasma proteins and the lysate is cooled to ambient temperature (15-25° C.) and filtered through a glass fiber or silica membrane. The filtrate is collected with a device like the one shown and saved. The first pass filtrate is mixed with 22 mL high stringency binding buffer (HSBB) to establish the high stringency binding condition, and filtered again through a glass fiber or silica membrane to capture the remaining DNA fragments. Each filter is washed and eluted to recover the 1.sup.st Pass, Large cfDNA Fraction and 2.sup.nd Pass, Small cfDNA Fraction. The twice filtered lysate is disposed of in waste.

(27) FIG. 27 shows a table of compositions for low and high stringency binding conditions for the methods depicted in FIG. 26 and FIG. 31, and tested throughout FIGS. 28 to 35. The upper table lists components and concentrations established with low stringency buffer (LSBB) when added to plasma or library products in the proportions described in FIGS. 26 and 31. The lower table lists components and concentrations established following addition of high stringency HSBB to lysates recovered following 1.sup.st Pass vacuum or 1.sup.st Spin Column filtration.

(28) FIG. 28 graphically depicts elution of spiked fragments of DNA size 72, 118, 194, 310, and 1078 base pairs from the 1st pass filtration (large DNA fragments) with lysates titrated with Tris buffer to a pH between 6.6 to 7.15 in the low stringency binding condition. As the pH of the low stringency binding condition increases, the affinity of the shorter DNA fragments for the glass fiber filter or silica membrane decreases. Reciprocally, as pH decreases the affinity of shorter DNA fragments for the glass fiber filter or silica membrane, increases. This differential affinity was achieved following the addition of high stringency binding buffer (HSBB) without changing the concentration of the chaotropic salt.

(29) FIG. 29 graphically depicts percent recovery of spike targets in the large (1.sup.st column

elution) and small (2.sup.nd column elution) from 10 mL of pregnancy plasma samples processed by Size Selection At Purification (SSAP). 200 μ L of 20 mg/mL Proteinase K and 11.7 mL of low stringency binding buffer (LSBB) was added to each 10 mL plasma to establish the low stringency binding condition and prepared larger fragments of DNA to selectively bind onto the glass fiber filter/silica membrane. In this example, the majority of DNA spike fragments 194 bp and larger were captured on the 1.sup.st pass filter. High stringency binding buffer (HSBB) was then added to the filtrate from the 1.sup.st pass filter to establish the high stringency binding condition where the smaller DNA fragments not bound by the 1.sup.st pass filter are efficiently captured by the 2.sup.nd pass glass fiber filter/silica membrane. This is revealed by the selective capture of the 72, 118, and some 194 bp, DNA spike fragments in the 2.sup.nd pass eluate presented for each case. The size selection obtained with this method shows exclusion of large DNA fragments in the small fraction which is directly correlated with the increased child fraction estimates (% CFE), and showing convincingly that the small fraction is enriched in fetal cfDNA.

(30) FIG. 30 graphically depicts the SSAP method as applied to 10 pregnancy plasma samples (the same as samples analyzed in FIG. 29) to obtain child fraction estimates (CFE) for each. The plots compare cfDNA fraction separated by Ctrl (Control) with no size selection, reflecting the total cfDNA fraction, 1.sup.st Elution (large cfDNA fraction), and 2.sup.nd Elution (small cfDNA fraction), for each case. Two $\times$ 10 mL plasma samples were extracted for each case. cfDNA from one 10 mL sample was processed to obtain all cfDNAs in one fraction (total cfDNA), while the second 10 mL sample was performed using the SSAP method to generate cfDNA from the 1.sup.st and 2.sup.nd filter pass. For every case, the 1.sup.st pass larger cfDNA fraction, enriched in maternal DNA, generated a lower CFE compared to the Ctrl. The 2.sup.nd pass small cfDNA fraction was enriched for fetal cfDNA fragments resulting in an increase in CFE compared to control.

(31) FIG. 31 shows the No Salt Increase-Library Size Selection (NSI-LSS) workflow. This workflow demonstrated dual-filtration size selection in a spin column format that has been applied to partition DNA library products into a large and a small fraction. This method utilized the same low and high stringency binding principle, wherein chaotropic salt in the low stringency binding condition begins at $>30\%$ and remained unchanged, or decreased slightly, in the high stringency binding condition. A purified or unpurified cfDNA library sample (186 μ L) is combined with 213 μ L of low stringency binding buffer (LSBB) to establish the low stringency binding condition, and the mixture is then applied to a glass fiber filter or silica membrane spin column and placed into a spin tube (as shown). The assembly is placed into a centrifuge for the 1.sup.st spin filtration, which allows the low stringency binding condition to make contact with the solid support, while the filtrate collects in the lower spin tube and is saved for secondary processing. The 1.sup.st spin column is removed and spin processed to wash $1\times$ with wash buffer and $1\times$ with EtOH, and elute in a low ionic strength buffer (i.e., 10 mM Tris, 0.1 mM EDTA, pH 8). This recovers a library fraction containing the largest of the DNA fragments (i.e., Large Fraction). The filtrate saved from the 1.sup.st spin is mixed with 400 μ L high stringency binding buffer (HSBB) to create the high stringency binding condition and applied to a new glass fiber or silica membrane spin column. The 2.sup.nd spin filtration allows the high stringency binding condition to contact the solid support and capture the remaining smaller DNA fragments. The 2.sup.nd spin column is removed, washed $1\times$ with wash buffer $1\times$ with EtOH, and eluted in TE to return the smallest of the DNA fragments (i.e., Small Fraction).

(32) FIG. 32 graphically depicts an electropherogram (CE) tracer generated by BioAnalyzer™ 1K of a DNA Ladder treated with the NSI-LSS method detailed in FIG. 31. Each CE trace exhibited size markers at 15 and 1500 base pairs, which flank peaks in the ZipRuler™ DNA ladder. The top panel shows peaks in the native ladder. The middle panel shows peaks eluted from the 1.sup.st spin column (Large DNA Fraction) following NSI-LSS separation. The bottom panel shows peaks eluted from the 2.sup.nd spin column (Small DNA Fraction). The Large DNA Fraction shows a

marked reduction in peaks at 200 and 300 bp in size and an absence of the 100 bp peak.

Conversely, the Small DNA Fraction shows recovery of all the 100 bp peak and the balance of the 200 and 300 bp peaks that were not captured by the 1.sup.st pass.

(33) FIG. **33** graphically depicts an electropherogram generated by BioAnalyzer™ 1K, resolving amplified library products size selected by the NSI-LSS method detailed in FIG. **31**. Library products eluting from the 1.sup.st spin column contained DNA fragments originating from cell-free DNA of di-nucleosomal and mono-nucleosomal in size. The library method adds 63 bp to each cfDNA fragment, so that the peaks observed at 239 and 391 bp as shown in the graph are close to the expected peak sizes of ~230 and ~380 bp for mono- and di-nucleosome derived fragments, respectively. Library products eluting from the 2.sup.nd spin column were devoid of fragments in the di-nucleosome size range (~380 bp). The peak labeled Small Fraction Library migrated with an average length of 229 bp, 10 bp shorter than the equivalent peak from the 1.sup.st spin column. Also, the small fraction appears to be enriched in even smaller fragments, as evidenced by the rise in the left shoulder of the graph, indicating the presence of library products originating from smaller cfDNA fragments.

(34) FIG. **34** graphically depicts percent recovery of spike fragments at 72, 118, 194, 310, and 1078 base pairs, that were added to libraries prepared from cfDNA derived from pregnancy plasma samples. Following library size selection by the NSI-LSS method (FIG. **31**), spike recovery of all 5 was determined by qPCR. The NSI-LSS method subdivided each library into a Large DNA fraction and a Small DNA fraction. The majority of 310 bp spike fragments were bound by 1st spin column and were therefore returned at a high rate of recovery (66-76%) within the Large Fraction Library. The 2.sup.nd spin column by contrast enriched the 72, 118, and 194 bp fragments in the Small Fraction Library, which are much smaller than 310 bp.

(35) FIG. **35** graphically depicts the NIS-LSS method as applied to amplified libraries generated from 13 pregnancy plasma samples to test the effect on child fraction estimates (CFE). The data are plotted to compare % CFE for control (Ctrl), having no size selection, and representing the total library DNA fraction, with the Large Library DNA Fraction (1.sup.st spin column elution), and the Small Library DNA Fraction (2.sup.nd spin column elution). Experimentally, a portion of non-selected library from each pregnancy case served as control. Another portion of the library was treated to NSI-LSS to obtain % CFE for library product returned from the 1.sup.st spin and 2.sup.nd spin columns. In every case but case #9, the control and large elution (1.sup.st spin column eluate) had lower % CFE than the library product eluted from the 2.sup.nd spin column. This demonstrates that smaller DNA fragments not captured by the 1.sup.st spin column are enriched for smaller fetal DNA fragments.

(36) FIG. **36** graphically depicts the LibAddition strategy to select for even shorter preserved cfDNA fragments without changing the size selection properties of the NSI-LSS method. The NSI-LSS method as it is applied in FIG. **31**, and demonstrated in FIGS. **32** and **34**, restricts library products <~310 bp in eluates from the 2.sup.nd spin column, (the Small Fraction Library). Under these conditions, cfDNA fragments <~250 bp in size (310-63=247) are enriched. To enrich for cfDNA fragments preserved in the library even smaller in size, without altering the chemistry of the method, one can append each fragment with a known length of DNA to shift the average size of the population of library products. FIG. **36** depicts such a strategy for adding a length of 25 bp to each library product by tailing PCR. As depicted, one round of PCR with primers F1/R1-LibAddition would add ~50 bp. Another ~50 bp would then be added to the F1/R1-LibAddition long product in a subsequent round of amplification with primers F2/R2-LibAddition. Under the scenario depicted, the cutoff for cfDNA fragments selected in the Small Library Fraction would be <~200 bp for F1/R1 products and <~150 bp for the F2/R2 products. Given that the underlying cfDNA fragments preserved in original libraries are ~166 bp, (incidentally in close agreement with data in FIG. **33** (~229-63=166 bp)). This means that size selection of the F1/R1-LibAddition products would shift the average cutoff to <~116 bp and further to a cutoff of <~66 bp if applied to the F2/R2-LibAddition

products.

DETAILED DESCRIPTION

(37) Reference will now be made in detail to some specific embodiments of the invention contemplated by the inventors for carrying out the invention. Certain examples of these specific embodiments are illustrated in the accompanying drawings. While the invention is described in conjunction with these specific embodiments, it will be understood that it is not intended to limit the invention to the described embodiments. On the contrary, it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims.

(38) In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present invention. Particular example embodiments of the present invention may be implemented without some or all of these specific details.

(39) Various techniques and mechanisms of the present invention will sometimes be described in singular form for clarity. However, it should be noted that some embodiments include multiple iterations of a technique or multiple instantiations of a mechanism unless noted otherwise.

(40) Introduction

(41) Characteristics of cfDNA in the circulation. The half-life of cfDNA can be longer than naked DNA spiked into fresh, unpreserved, plasma or when injected into the bloodstream in vivo. This can be due to the fact that circulating nuclear DNA remains in tight association with core and linker histones which protect two wraps or gyres of DNA, in mononucleosomes and chromatosomes, from active nucleases in blood or plasma, thus preserving fragments of ~130 to ~170 base pairs (bp) in length. Fragments of two or three times this length can also be recovered from plasma, demonstrating that oligonucleosomes and oligochromatosomes can exist in the circulation as well. In addition to chromatinized DNA, both DNA and various RNA species survive for a substantial length of time in the circulation within membrane bound microvesicles (exosomes), actively shed by cells via exocytosis and blebbing. The steady-state concentration of circulating cell free DNA (cfDNA) fluctuates in the ng/mL range, and reflects the net balance between release of fragmented chromatin into the bloodstream and the rate of clearance by nucleases, hepatic uptake and cell mediated engulfment. Normal and health compromised individuals, exhibit cfDNA concentrations averaging 1 and 40 ng/mL of plasma (*J. Clin. Inv.* (1975) 56:512). No single source or mechanism can explain from where or how such short chromatin bits enter the circulation with such regularity, but as discussion, the process is dominated by erythrocytic apoptosis in the blood and bone marrow. Lesser contributions from apoptotic, necrotic and traumatic cell death, coupled with macrophage destruction throughout the body (*Cancer. Res.* (2001) 61:1659) spill cfDNA sequences into the blood that potentially include rare variants indicative of latent disease or serious fetal genetic anomalies. When coupled to the power of next generation genetic testing, cfDNA can provide unprecedented access to genetic information from disease states that might elude conventional detection, or where the site of origin is inaccessible to biopsy. Accurate and early detection of tumor associated genetic mutation, rearrangements, copy number variation, insertions/deletions or fusions is possible through deep analysis of cfDNA from plasma.

(42) Preservation of cfDNA for genetic analysis. The key to liquid biopsy approaches which target cfDNA, is the ability to bind and purify sufficient quantities of the highly fragmented DNA from blood plasma collected by needle stick, typically from an arm vein. With respect to non-invasive prenatal testing and cancer detection, a huge problem is presented by the fact that an overwhelming majority of cfDNA in blood comes from normal cells. This background of normal DNA dilutes the far scarcer fragments originating from the developing fetus or tumor cells. Thus care needs to be taken to preserve circulating nucleosomes from the time of blood collection to sample processing, and to prevent or minimize further dilution of cfDNA by genomic DNA released by lysis of nucleated cells. Such precautions begin at blood collection with the utilization of blood collection tubes (BCT's) which contain anticoagulating and cell stabilizing agents which prevent lysis of

mononuclear cells during storage for up to 14 days. To compensate for the low endogenous levels of cfDNA in plasma and to improve the odds of sampling a comparatively rare population of sequences of interest, tests routinely call for the processing of large volumes, up to 10 mL, of plasma through DNA extraction methods. This necessitates collection of at least two 10 mL blood samples to generate one 10 mL plasma sample. The present invention describes methods for release of bound cfDNA from nucleoprotein complexes contained in human plasma and the high efficiency capture and recovery (>85-95%) of the liberated cfDNA fragments from 10 mL of plasma. The method is extendable to isolation of cfDNA from serum and other body fluids.

(43) DNA Extraction from large volume plasma samples. The isolation and purification of cfDNA from plasma poses a particular set of challenges due to the low starting concentration, matrix complexities, and the variable nature of plasma samples collected by venipuncture into vacuum tubes. Conventionally, 10 to 60 ng of cfDNA is recoverable from 10 mL of human plasma, and the average small size of DNA fragments make them difficult to capture and retain on solid supports through sequential wash steps. Plasma is a complicated fluid, and in comparison to the total mass of other macromolecular constituents (e.g., proteins, lipids and protein-lipid complexes), cfDNA represents a tiny fraction. Any successful plasma nucleic acid extraction process needs to accomplish three things to isolate cfDNA in pure form and at high rates of recovery. First, the protein complexes that serve to protect cfDNA (i.e., chromatinized DNA in the form of mono-, di-, tri-nucleosomes or longer) from nucleases need to be deconstructed to release cfDNA and expose it for capture on solid phases. Second, the macromolecular components which predominate in plasma (e.g., albumin, immunoglobulins, fibrinogen/fibrin, free hemoglobin, proteinase inhibitors, nucleases, lipids and lipoprotein complexes) need to be dissolved, degraded, solubilized, or neutralized to prevent them from interacting with released cfDNA or the capture matrix in ways that would interfere with (for example clog or foul) or reduce the efficiency of nucleic acid binding. Third, the establishment of a chemical environment, binding proficient condition or nucleic acid binding state that supports and promotes complete, preferential, stable, and reversible interaction of nucleic acids, in particular cfDNA fragments of all sizes, with the solid phase support material or capture matrix comprised of glass fiber or silica.

(44) Release of cfDNA by proteolysis, chemical denaturation or both. The two main methods used to disrupt stable noncovalent DNA-protein interactions are chemical denaturation and enzymatic destruction. Early methods employed organic liquid phase extraction utilizing phenol and phenol-chloroform mixtures to disintegrate nucleoprotein complexes and sequester proteins and lipids into the organic phase while partitioning the highly hydrophilic DNA and RNA into the aqueous phase in very pure form. Phenol-chloroform methods proved highly efficient and delivered DNA highly suitable for enzymatic manipulation. However, user and environmental safety, ease of use considerations, and practical difficulties of scaling large volume extractions to phenol-chloroform methods have led to its replacement with safer, highly scalable solid phase methods that can more easily purify nucleic acids from almost any starting material. One of the earliest solid phase methods used to purify DNA was described by E.M. Southern (*J. Mol. Biol.* (1975) 94:51-70) where the DNA excised from agarose hydrogels was recovered following dissolution in strongly chaotropic salts, sodium perchlorate or sodium iodide (NaI), followed direct DNA capture on hydroxyapatite (mineralized calcium phosphate) particles, washed and eluted into a low ionic strength buffer. Vogelstein and Gillespie (*PNAS, USA* (1979) 76:615-619) later improved upon this earliest example by substituting powdered glass for hydroxyapatite and captured DNA from bits of agarose gels dissolved in saturated NaI. Excess NaI was removed by washing glass particles in 50% buffered ethanol and the bound DNA eluted in Tris buffered saline, EDTA. This method, which utilized glass or silica as a solid support to bind nucleic acids in the presence of high salt, followed by washes in high percentage alcohol to remove contaminants, and elution in low ionic strength buffers, forms the basis for most commercial nucleic acid purification kits on the market. These safer and highly scalable methods work by exploiting the strong yet reversible hydrophilic

interaction promoted between DNA and silanols and siloxanes on the surface of glass and silica (*Colloids and Surfaces, A: Physicochemical and Engineering Aspects*, (2000)173:1-38) in high salt solutions. Unlike phenol-chloroform methods which efficiently denature and strip bound proteins off DNA and simultaneously denature, solvate and move proteins, lipids and other contaminants into the organic phase, solid phase extraction methods need to deal with DNA bound proteins and background sample contaminants differently. Proteolysis of protein-DNA complexes is the most widely employed method of releasing proteins bound to DNA and for degrading other protein contaminants contained in the starting sample. Still other effective methods utilize only strong chemical denaturants to disrupt protein tertiary and secondary structure, dissociate DNA/RNA from chromatin or binding proteins, and unfold other proteins contained in the sample to greatly diminish their interference with the glass/silica solid phase during DNA capture. Boom et al. (*J Clin Micro.* (1990) 28(3):495-503) were the first to detail the use of solid phase capture on powdered glass and diatomaceous silica from clinical samples such as serum and urine. Their method used a solid phase of glass or silica particles to adsorb nucleic acids from complex biological samples following direct chemical lysis in high concentrations of chaotropic salts.

(45) A generalized scheme by which cfDNA can be isolated from plasma is presented in FIG. 8, which describes major effectors for each phase of the extraction. For cfDNA isolation by solid phase capture, plasma proteins and protein-DNA complexes are typically disrupted by a combination of proteolytic lysis and chemical lysis which sets up the nucleic acid binding state, a condition that necessitates the sequential, serial addition of two buffers, Proteolysis Buffer and Binding Buffer to samples, separated by an incubation step (see FIG. 8, steps 2 & 3). The constituents of Proteolysis Buffer and Binding Buffer should be optimized to effect complete proteolysis and the combination of which should establish a chemical environment that promotes highly efficient interaction of nucleic acids (DNA/RNA) with a Solid Phase or Binding Matrix (FIG. 8, step 4) such as glass fiber or silica particles. Proteinase K is the most common broad spectrum protease used for proteolytic lysis in DNA extraction methods. It is a stable serine protease that is active under a wide range of pH, temperature, salt, solvent, and detergent concentrations. The activity of Proteinase K peaks in the presence of moderate denaturants, 2-4 molar chaotropic salts and ionic detergents, which act both to stimulate enzymatic activity and increase substrate accessibility by destabilizing protein secondary structure. At completion, Proteinase K digestion will have reduced polypeptides to small di- and tri-peptides, and in the process degraded itself by autodigestion, thus eliminating the vast majority of enzyme added to samples. Proteolysis Buffer is a key additive in DNA extraction methods, and critical to DNA isolation from complex biological samples. In sample mixtures, Proteolysis Buffer is designed to preserve target nucleic acids, establish optimum conditions for proteolysis, solubilize lipids and microvesicles, breakdown colloids and particulate matter, and prevent precipitation over the course of protease reactions. Moreover, Proteolysis Buffer must be compatible with Binding Buffers which are added to samples following proteolysis in order to complete the denaturation process and establish the nucleic acid binding state (see FIG. 8, step 3). Binding Buffers act to chemically complete denaturation, quench remaining PK activity, and sets up a nucleic acid binding state that ensures high efficiency capture of short nucleic acids to silicate supports (see FIG. 8, steps 3 & 4). Available methods designed to isolate cfDNA from plasma or serum typically begin with a proteinase K lysis step initiated under moderately harsh conditions optimized for protease activity, followed by much harsher and highly denaturing chemical lysis steps optimized for DNA binding. Proteolysis Buffers and Binding Buffers serve two separate yet complimentary functions when combined with sample matrices in an ordered fashion, and form an articulated chemical system that supports high level solid phase adsorption of large and small nucleic acid fragments contained in complex biological samples.

(46) Many next generation genetic tests utilize plasma cfDNA from a simple blood draw as an input. This patient sampling technique known as a liquid biopsy is considered a non-invasive

medical procedure valuable in cancer surveillance (*J Clin Oncol.* (2014) 32(6):579-586) and detection, and prenatal health screening (*Annu Rev Genomics Hum Genet.* (2012)13:285-306). Non-invasive prenatal tests (NIPT's) which utilize cfDNA from the plasma of pregnant women to detect chromosomal aneuploidies and microdeletions that may affect child health, are prime examples of such liquid biopsy based NGS tests. Most NGS assays begin with the preservation and amplification of the very small amounts of cfDNA obtained from plasma samples in a process known as library preparation. Construction of the library immortalizes the original cfDNA isolate and uniformly multiplies the sample through a series of molecular reactions that enzymatically repair, tail, and amplify fragments to prepare them for NGS analysis. In the NIPT assay referred to herein, libraries are subject to massively multiplexed amplification reactions that amplify single nucleotide polymorphisms (SNPs) used in the genetic analysis. The amplified SNP targets are then barcoded and readied for NGS sequencing. Sequence data is processed and allelic designations for each SNP are assigned to the mother or fetus (i.e., of paternal origin) according to a bimodal mixture model of homozygous (AA) or heterozygous (AB) allele distribution (*Bioinformatics*, 28(2):2883-2890). A higher fraction of fetal cfDNA in plasma isolates leads to a greater proportion of fetal SNP's out of the total (maternal+fetal) for each target SNP detected. A higher fetal fraction produces a greater divergence between the fetal genotype and the underlying maternal genotype, and thus increases the call confidence of ploidy estimates at the chromosome and locus level. More than one factor can profoundly influence the fetal fraction in cfDNA preparations, most critical is the storage condition and anticoagulant preservative used in blood collection tubes and the time between collection and plasma isolation. Conditions that minimize lysis of leucocytes significantly reduces leakage of maternal genomic DNA into the plasma, and thereby increase the fraction of fetal cfDNA as a percentage of total. Additionally, DNA purification methods that recover the broadest range of DNA sizes, particularly small fragments <100 bp in length, will ensure yield of the highest fetal fraction. This derives from the fact that circulating fetal DNA is on average ~23 bp shorter (143 bp vs 166 bp) than maternal cfDNA (PNAS, USA (2016) 113(50) E8159-E8168). Most recent evidence, based on the analysis of ssDNA libraries, suggests that much more cfDNA shorter in length is present (*Cell* (2016) 164:57-68), but indeed much of it may be excluded by the extraction method and library construction processes themselves (PNAS, USA (2016) 112(11):3178-3179). Thus plasma cfDNA extraction methods that rescue short <100 bp, <75 bp, or even <50 bp cfDNA fragments may well be expected to return higher fetal fraction estimates than methods that do not. FIG. 9 compares the fetal fraction estimates from 16 paired maternal samples where plasma cfDNA was isolated with IPA or ACN used as the co-solvent to establish the nucleic acid binding state. A highly statistically significant increase in the average fetal fraction was obtained from the otherwise identical analysis treatment of the cfDNA isolated with acetonitrile compared to isopropanol. This result is surprising and it was not anticipated that an increase in fetal fraction would result from the substitution of a protic solvent such as IPA with the aprotic solvent ACN. Though highly unexpected, the increase in fetal fraction could be explained by an improve preservation and subsequent recovery of short cfDNA fragments.

(47) Two-Step Filtration Method for Capturing cfDNA

(48) Many embodiments described herein relate to a method for isolating nucleic acids from a biological sample, comprising: (a) contacting a first composition comprising nucleic acids obtained from a biological sample with a first matrix under a low-stringency binding condition in the presence of <1% aliphatic alcohols, that binds less than 5% of nucleic acids of about 72 bp and more than 30% of nucleic acids of about 194 bp to the first matrix; and (b) contacting a second composition comprising the remainder of the first composition with a second matrix under a high-stringency binding condition at less than 1% aliphatic alcohol, that binds more than 70% of nucleic acids of about 72 bp and more than 30% of nucleic acids of about 50 bp to the second matrix.

(49) In some embodiments, step (a) comprises filtering the first composition through the first matrix to obtain a filtrate, and the second composition comprises the filtrate of step (a). In some

embodiments, after nucleic acids are bound to the first matrix, the method further comprises washing the first matrix with a washing buffer, drying the matrix, and/or eluting nucleic acids from the first matrix with an elution buffer.

(50) In some embodiments, step (b) comprises filtering the second composition through the second matrix. In some embodiments, after nucleic acids are bound to the second matrix, the method further comprises washing the second matrix with a washing buffer, drying the matrix, and/or eluting nucleic acids from the second matrix with an elution buffer.

(51) In some embodiments, the method further comprises incubating the biological sample with a protease such as proteinase K prior to step (a). The biological sample can be, for example, a sample of a maternal blood, plasma, or serum. The biological sample can be, for example, a plasma sample from a pregnant woman comprising fetal cfDNA and maternal cfDNA, or a plasma sample from a cancer patient comprising circulating tumor DNA. In addition, the biological sample can comprise cfDNA selected from, for example, nucleic acids of virus, fungal or bacterial origin, as virus or virus-like particles, fungal mycelium, yeast or bacterial cells, in particle-free, cell-free, aggregate, vesicle or platelet bound forms.

(52) In some embodiments, the first composition of step (a) is obtained by adding a first binding buffer to the biological sample after digestion by protease, wherein the first binding buffer establishes the low-stringency binding condition.

(53) In some embodiments, the second composition of step (b) is obtained by adding a second binding buffer to the filtrate of step (a), wherein the second binding buffer establishes the high-stringency binding condition.

(54) In some embodiments, the first and/or second binding buffer comprises a chaotropic compound and a solvent, wherein the solvent comprises a nitrile compound, tetrahydrofuran (THF), or a combination thereof.

(55) In some embodiments, the first and/or second binding buffer comprises a nitrile compound selected from acetonitrile (ACN), propionitrile (PCN), butyronitrile (BCN), isobutylnitrile (IBCN), or a combination thereof. The first and/or second binding buffer can comprise, for example, about 15% to about 35%, or about 20% to about 30%, or about 25% of the nitrile compound (e.g., ACN).

(56) In some embodiments, the first and/or second binding buffer comprises a chaotropic compound selected from GnCl , urea, thiourea, guanidine thiocyanate, NaI , guanidine isothiocyanate, D-/L-arginine, a perchlorate or perchlorate salt of Li^+ , Na^+ , K^+ , or a combination thereof. The first and/or second binding buffer can comprise, for example, about 5 M to about 8 M, or about 5.6 M to about 7.2 M, or about 6 M of the chaotropic compound (e.g., GnCl).

(57) After the addition of the first binding buffer, the first composition can comprise, for example, about 4% to about 6%, or about 4.8% to about 5.6% of the nitrile compound (e.g., ACN). The first composition can also comprise, for example, about 3 M to about 4 M, or about 3.2 M to about 3.4 M of the chaotropic compound (e.g., GnCl).

(58) After the addition of the second binding buffer, the second composition can comprise, for example, about 10% to about 20%, or about 13% to about 18%, or about 14% to about 15% of the nitrile compound (e.g., ACN). The second composition can also comprise, for example, about 3.5 M to about 6 M, or about 4 M to about 5 M, or about 4.3 M to about 4.5 M of the chaotropic compound (e.g., GnCl).

(59) The pH of the first binding buffer can be, for example, about 8 to about 10, or about 8.5 to about 9.5, or about 8.9 to about 9.1, or about 9. The pH of the second binding buffer can be, for example, about 4 to about 6, or about 4.5 to about 5.5, or about 4.9 to about 5.1, or about 5. The pH of the first and/or second binding buffer can be adjusted using a buffering agent (e.g. MES or 2-(N-morpholino)ethanesulfonic acid, Tris(hydroxymethyl) aminomethane (Tris-base), or a mono, di- or tri-carboxylic acid such as formic, acetic, malonic, succinic, glutaric, citric, or malic).

(60) In some embodiments, after the addition of the first binding buffer, the first composition can have a pH of, for example, about 7 to about 10, about 6 to about 8, or about 6 to about 7, or about

6.2 to about 6.8, or about 6.3 to about 6.7. In some embodiments, after the addition of the second binding buffer, the second composition can have a pH of, for example, about 4 to about 6, or about 5 to about 6, or about 5.3 to about 5.9, or about 5.4 to about 5.8.

(61) In some embodiments, the first and/or second binding buffer comprises less than 5% of alcohol, or less than 2% of alcohol, or less than 1% of alcohol, or less than 0.1% of alcohol, or comprises no alcohol. In some embodiments, the first and/or second binding buffer comprises less than 5% of propanol, or less than 2% of propanol, or less than 1% of propanol, or less than 0.1% of propanol, or comprises no propanol such as isopropanol. In some embodiments, the first and/or second binding buffer comprises less than 5% of non-water protic solvents, or less than 2% of non-water protic solvents, or less than 1% of non-water protic solvents, or less than 0.1% of non-water protic solvents, or comprises no non-water protic solvents.

(62) Alternatively, in some embodiments, the first and/or second binding buffer comprises an alcohol solvent instead of a nitrile solvent such as ACN. In some embodiments, the first and/or second binding buffer comprise isopropanol (IPA).

(63) In some embodiments, the matrix comprises siliceous materials, silica gel, glass, glass fiber, zeolite, aluminum oxide, titanium dioxide, zirconium dioxide, kaolin, gelatinous silica, magnetic particles, ceramics, polymeric supporting materials, or a combination thereof. In a particular embodiment, the matrix comprises glass fiber. In one embodiment, the first matrix is different from the second matrix. In another embodiment, the first matrix is the same as the second matrix.

(64) In some embodiments, the first composition and/or the second composition further comprises a chelating compound. The chelating compound can be, for example, ethylenediaminetetracetic (EDTA), ethyleneglycol-bis(2-aminoethylether)-N,N,N',N'-tetraacetic acid (EGTA), citric acid, N,N,N',N'-Tetrakis(2-pyridylmethyl)ethylenediamine (TPEN), 2,2'-Bipyridyl, deferoxamine methanesulfonate salt (DFOM), 2,3-Dihydroxybutanedioic acid (tartaric acid), or a combination thereof. In a particular embodiment, the chelating compound is EDTA.

(65) In some embodiments, the first composition and/or the second composition further comprises a detergent. The detergent can be, for example, Triton X-100, Tween 20, N-lauroyl sarcosine, sodium dodecylsulfate (SDS), dodecyltrimethylphosphine oxide, sorbitan monopalmitate, decylhexaglycol, 4-nonylphenyl-polyethylene glycol, or a combination thereof. In a particular embodiment, the detergent is Triton X-100.

(66) In some embodiments, the first binding buffer and/or the second binding buffer can comprise, for example, about 1% to about 6% of Triton X-100, or about 2% to about 4% of Triton X-100, or about 3% of Triton X-100. In some embodiments, after the addition of the first binding buffer, the first composition can comprise, for example, about 5% to about 6% of Triton X-100, or about 5.3% to about 5.6% of Triton X-100, or about 5.4 to about 5.5% of Triton X-100. In some embodiments, after the addition of the second binding buffer, the second composition can comprise, for example, about 4% to about 5% of Triton X-100, or about 4.2 to about 4.5% of Triton X-100, or about 4.3% to about 4.4% of Triton X-100.

(67) The low stringency binding condition of step (a) is configured to restrict binding of shorter nucleic acid molecules while permitting binding of longer nucleic acid molecules to the matrix or solid support. In some embodiments, step (a) comprises binding less than 20%, or less than 10%, or less than 5%, or less than 2%, or less than 1% of nucleic acids of about 72 bp to the first matrix under the low-stringency binding condition. In some embodiments, step (a) comprises binding less than 70%, or less than 60%, or less than 50%, or less than 40%, or less than 30%, or less than 20%, or less than 10%, or less than 5% of nucleic acids of about 118 bp to the first matrix under the low-stringency binding condition. In some embodiments, step (a) comprises binding less than 90%, or less than 80%, or less than 70%, or less than 60%, or less than 50%, or less than 40%, or less than 30%, or less than 20% of nucleic acids of about 194 bp to the first matrix under the low-stringency binding condition. In some embodiments, step (a) comprises binding more than 20%, or more than 30%, or more than 40%, or more than 50%, or more than 60%, or more than 70%, or more than

80%, or more than 90% of nucleic acids of about 1078 bp to the first matrix under the low-stringency binding condition.

(68) In contrast, the high stringency binding condition of step (a) is configured to facilitate binding of shorter nucleic acid molecules to the matrix or solid support. In some embodiments, step (b) comprises binding more than 40%, or more than 50%, or more than 60%, or more than 70%, or more than 80%, or more than 90% of nucleic acids of about 72 bp to the second matrix under the high-stringency binding condition. In some embodiments, step (b) comprises binding more than 40%, or more than 50%, or more than 60%, or more than 70%, or more than 80%, or more than 90% of nucleic acids of about 118 bp to the second matrix under the high-stringency binding condition. In some embodiments, step (b) comprises binding more than 40%, or more than 50%, or more than 60%, or more than 70%, or more than 80%, or more than 90% of nucleic acids of about 194 bp to the second matrix under the high-stringency binding condition. In some embodiments, step (b) comprises binding more than 50%, or more than 60%, or more than 70%, or more than 80%, or more than 90% of nucleic acids of about 1078 bp to the second matrix under the high-stringency binding condition. In some embodiments, step (b) comprises binding more than 20%, or more than 30%, or more than 40%, or more than 50%, or more than 60% or more than 70% of nucleic acids of about 50 bp to the second matrix under the high-stringency binding condition.

(69) In some embodiments, the fetal fraction of nucleic acids elutable from the second matrix is at least 1%, or at least 2%, or at least 3%, or at least 5%, or at least 10% higher than fetal fraction of nucleic acids elutable from the first matrix. In some embodiments, the fetal fraction of nucleic acids elutable from the second matrix according to the two-step filtration method described herein (see e.g., FIG. 14) is at least 0.5%, or at least 0.8%, or at least 1%, or at least 2%, or at least 3%, or at least 5% higher than fetal fraction of nucleic acids elutable from a corresponding matrix according to the single-step filtration method (see e.g., FIG. 13).

(70) In some embodiments, the method described herein does not comprise molecular sieving and chromatographic techniques. In some embodiments, the method described herein does not comprise covalent modification, conjugation, or labeling of the nucleic acids. In some embodiments, the method described herein does not comprise differential adsorption to solid supports such as Solid Phase Reversible Isolation (SPRI) selection. In some embodiments, the method described herein does not comprise use of artificial crowding agent such as polyethylene glycol (PEG), ficoll, dextran, or serum albumin.

(71) Further embodiments described herein relate to a kit for isolating nucleic acids from a biological sample, comprising (a) the first binding buffer described herein for establishing the low-stringency binding condition for binding nucleic acids to the first matrix, and (b) the second binding buffer described herein for establishing the high-stringency binding condition for binding nucleic acids to the second matrix. In some embodiments, the kits further comprises a digestion buffer, a protease, a washing buffer, and/or an elution buffer.

(72) Further Embodiments of Binding Composition and Binding Buffer

(73) Many embodiments described herein relate to a composition for isolating nucleic acids from a biological sample, comprising a chaotropic compound and a solvent, wherein the solvent comprises an aprotic solvent such as a nitrile compound, tetrahydrofuran, or a combination thereof.

(74) In some embodiments, the solvent comprises a nitrile compound. The nitrile compound can be, for example, acetonitrile (ACN), propionitrile (PCN), butyronitrile (BCN), isobutylnitrile (IBCN), or a combination thereof.

(75) In a particular embodiment, the nitrile compound is ACN. The composition can comprise, for example, about 10% to about 20% of ACN, or about 13% to about 18% of ACN, or about 15% of ACN.

(76) In some embodiments, the composition comprises less than 10%, or less than 5%, or less than 2%, or less than 1% of alcohol, or substantially or totally free of alcohol. In some embodiments, the composition comprises less than 10%, or less than 5%, or less than 2%, or less than 1% of propanol

such as isopropanol, or substantially or totally free of isopropanol. In some embodiments, the composition comprises less than 10%, or less than 5%, or less than 2%, or less than 1% of non-water protic solvents, or substantially or totally free of non-water protic solvents. The pH of the composition can be, for example, about 4 to about 10, or about 4 to about 5, or about 5 to about 6, or about 6 to about 7, or about 7 to about 8, or about 8 to about 9, or about 9 to about 10, or about 4 to about 8, or about 4.5 to about 6, or about 4.9 to about 5.1.

(77) The chaotropic compound can be, for example, guanidine chloride (GnCl), urea, thiourea, guanidine thiocyanate, NaI, guanidine isothiocyanate, arginine, hydrogen perchlorate or perchlorate salt of Li⁺, Na⁺, K⁺, or a combination thereof.

(78) In a particular embodiment, the chaotropic compound is GnCl. The composition can comprise, for example, about 2.0 M to about 3.5 M, about 3.5 M to about 6 M of GnCl, or about 4 M to about 5 M of GnCl, or about 4.4 M of GnCl.

(79) In some embodiments, the composition further comprises a chelating compound. The chelating compound can be, for example, ethylenediaminetetracetic (EDTA), ethyleneglycol-bis(2-aminoethylether)-N,N,N',N'-tetraacetic acid (EGTA), citric acid, N,N,N',N'-Tetrakis(2-pyridylmethyl)ethylenediamine (TPEN), 2,2'-Bipyridyl, deferoxamine methanesulfonate salt (DFOM), 2,3-Dihydroxybutanedioic acid (tartaric acid), or a combination thereof. In a particular embodiment, the chelating compound is EDTA.

(80) In some embodiments, the composition further comprises a detergent. The detergent can be, for example, Triton X-100, Tween 20, N-lauroyl sarcosine, sodium dodecylsulfate (SDS), dodecyltrimethylphosphine oxide, sorbitan monopalmitate, decylhexaglycol, 4-nonylphenyl-polyethylene glycol, or a combination thereof.

(81) In a particular embodiment, the detergent is Triton X-100. The composition can comprise, for example, about 3% to about 6% of Triton X-100, or about 4% to about 5% of Triton X-100, or about 4.5% of Triton X-100.

(82) In a particular embodiment, the first binding buffer (also referred to as low-stringency binding buffer or LSBB) comprises Tris-base, a nitrile compound, a chelating compound, a detergent, and a chaotropic compound. In a further embodiment, the first binding buffer comprises Tris-base in an amount of about 10 mM to about 50 mM, the nitrile compound in an amount of about 0% to about 20%, the chelating compound in an amount of about 0 mM to about 2 mM, the detergent in an amount of about 0% to about 15%, and the chaotropic compound in an amount of about 4 M to about 8 M. In a yet further embodiment, the nitrile compound is acetonitrile, the chelating compound is EDTA, the detergent is Tween 20, and the chaotropic compound is guanidine chloride. In one preferred embodiment, the LSBB comprise between about 0 to about 20% Acetonitrile, about 0 to about 15% Tween 20, about 6 to about 7.8 molar Guanidine chloride, about 10 to about 50 mM Tris (free base), about 0 to about 2 mM EDTA (free acid), and have a pH of about 7.0 to about 10.

(83) In a particular embodiment, the second binding buffer (also referred to as high-stringency binding buffer or HSBB) comprises MES, a nitrile compound, a detergent, an alcohol, a chaotropic compound, and a chelating compound. In a further embodiment, the second binding buffer comprises MES in an amount from about 10 mM to about 70 mM, the nitrile compound in an amount of about 10% to about 40%, the detergent in an amount from about 0% to about 10%, the alcohol in an amount of about 0 to about 2%, the chaotropic compound in an amount of about 2 M to about 4 M, and the chelating compound in an amount from about 0 mM to 2 mM. In one preferred embodiment, the HSBB may comprise between about 10 to about 40% Acetonitrile, about 0 to about 10% Tween 20, about 0 to about 2% Ethanol, about 3.2 to about 3.6 molar Guanidine chloride, about 10 to about 70 mM MES (free acid), about 0 to about 2 mM EDTA (free acid), and have a pH of about 4.05 to about 6.5.

(84) In a particular embodiment, contacting a biological sample with a first binding buffer forms the low-stringency binding condition. In one particular embodiment, the low-stringency binding

condition comprises the biological sample in an amount from about 40% to about 50%, the Tris-base in an amount from about 0.1% to about 0.3%, the chaotropic compound in an amount from about 25% to about 35%, the chelating compound from about 0% to about 0.05%, the detergent in an amount from about 0% to about 10%, and the nitrile compound in an amount from about 0% to about 10%. In one embodiment, the first binding buffer and/or the low-stringency condition has a pH of about 7.0 to 10. In one preferred embodiment, the low-stringency binding condition comprises one or more of the following: plasma (about 38.9% to about 45.66%), Acetonitrile (about 0% to about 12.1%), Tris base (about 5.43 mM to about 20 mM), Guanidine chloride (about 3.2055 M to about 4.76 M), Tween 20 (about 0% to about 9%), EDTA (about 0% to about 1.2%) and a pH of about 7 to about 9.

(85) In a particular embodiment, contacting a biological sample with the second binding buffer forms the high-stringency binding condition. In one particular embodiment, the high-stringency binding condition comprises the biological sample in an amount from about 40% to about 50%, the Tris-base in an amount from about 0.001% to about 0.5%, the MES in an amount from about 0.1% to about 1.0%, the chaotropic compound in an amount from about 25% to about 35%, the chelating compound from about 0% to about 0.05%, the detergent in an amount from about 0% to about 10%, the alcohol in an amount from about 0.1% to about 1%, and the nitrile compound in an amount from about 10% to about 30%. In one embodiment, the second binding buffer and/or the high-stringency binding condition has a pH of about 7.0 to 10. In one preferred embodiment, the high-stringency binding condition one or more of the following: plasma (about 19.27% to about 22.78%), Acetonitrile (about 4.59% to about 26.09%), Guanidine chloride (about 3.2055 M to about 4.76 M, optionally with concentrations holding constant at low and high stringency), Tween 20 (about 0 to about 9.55%), Tris-base (about 1.22 mM to about 5.33 mM), MES (about 4.59 mM to about 38 mM), Ethanol (about 0 to about 1%), EDTA (about 0 to about 0.61 mM), and a pH of about 4.05 to about 5.5.

(86) In some embodiments, the composition further comprises nucleic acids. The nucleic acids can comprise, for example, DNAs and/or RNAs.

(87) The nucleic acids can comprise, for example, maternal nucleic acids or fetal nucleic acids. The nucleic acids can comprise, for example, cell free nucleic acids or circulating tumor nucleic acids. The cell free nucleic acids may be obtained from a sample of a maternal blood, plasma, or serum. The nucleic acids can comprise, for example, DNAs or RNAs. The cell free nucleic acids can comprise, for example, cell free fetal DNA and cell free maternal DNA. The cell free DNA can comprise, for example, nucleic acids of virus, fungal or bacterial origin, as virus or virus-like particles, fungal mycelium, yeast or bacterial cells, in particle-free, cell-free, aggregate, vesicle or platelet bound forms.

(88) The nucleic acids can be, for example, about 50 to about 1200 base pairs in length, or about 70 to about 500 base pairs in length, or about 100 to about 200 base pairs in length, or about 130 to about 170 base pairs in length.

(89) In one particular embodiment, the cell-free DNA or circulating tumor DNA in the sample may be amplified prior to forming the low stringency binding condition. The amplification may be performed by ligating the cfDNA or the ctDNA to a plurality of DNA adapter molecules, wherein the DNA adapter molecules comprises common forward and reverse primer binding sites, and then amplifying the ligated cfDNA or ctDNA by using forward and reverse primers complementary to the common primer binding sites in the DNA adaptor molecules.

(90) In one particular embodiment, the size of the cell-free DNA or circulating tumor DNA in the sample may be increased with trailing PCR prior to forming the low-stringency binding condition. The cell-free DNA or circulating tumor DNA may be increased with 10 bp, 15 bp, 20 bp, 25 bp, 30 bp, or 35 bp.

(91) In some embodiments, the composition further comprises a matrix. The matrix can comprise, for example, siliceous materials, silica gel, glass, glass fiber, zeolite, aluminum oxide, titanium

dioxide, zirconium dioxide, kaolin, gelatinous silica, magnetic particles, ceramics, polymeric supporting materials, or a combination thereof. In a particular embodiment, the matrix comprises glass fiber.

(92) It was surprising and highly unexpected that such highly efficient recovery of nucleic acids, in particular cfDNA from plasma, could be achieved when protic solvents such as ethanol, propanol, or isopropanol were replaced by the aprotic solvents of the nitrile series including acetonitrile ((ACN), ethyl nitrile or methyl cyanide), propionitrile ((PCN), propyl nitrile or ethyl cyanide), butyronitrile ((BCN) butane nitrile or propyl cyanide), and isobutylnitrile ((IBCN), isobutyl nitrile or isopropyl cyanide), in the presence of a chaotropic compound through binding to a matrix such as glass fiber or silica. Just as unexpected was the fact that this combination also increased the calculated fetal fraction deriving from a SNP based NIPT method, given that contact times between the glass fiber matrix and the DNA binding state were much shorter than under binding conditions established with IPA as a solvent.

(93) Further embodiments described herein relate to a method for binding nucleic acids to a matrix and isolating the nucleic acids, comprising contacting the nucleic acids from a biological sample with a matrix in the presence of a chaotropic compound and a solvent, thereby binding the nucleic acids to the matrix, wherein the solvent comprises an aprotic solvent such as a nitrile compound, tetrahydrofuran, or a combination thereof.

(94) In some embodiments, the nucleic acids are contacted with the matrix in the presence of a nitrile compound selected from ACN, PCN, BCN, IBCN, or a combination thereof. In a particular embodiment, the nitrile compound is ACN. The nucleic acids can be contacted with the matrix in the presence of, for example, about 10% to about 20% of ACN, or about 13% to about 18% of ACN, or about 15% of ACN.

(95) In some embodiments, the nucleic acids are contacted with the matrix in the presence of less than 10% of alcohol, or less than 5% of alcohol, or less than 2% of alcohol, or less than 1% of alcohol, or substantially or totally in the absence of alcohol. In some embodiments, the nucleic acids are contacted with the matrix in the presence of less than 10% of propanol, or less than 5% of propanol, or less than 2% of propanol, or less than 1% of propanol such as isopropanol, or substantially or totally in the absence isopropanol. In some embodiments, the nucleic acids are contacted with the matrix in the presence of less than 10% of non-water protic solvents, or less than 5% of non-water protic solvents, or less than 2% of non-water protic solvents, or less than 1% of non-water protic solvents, or substantially or totally in the absence non-water protic solvents.

(96) In some embodiments, the nucleic acids are contacted with the matrix in the presence of a chaotropic compound selected from GnCl, urea, thiourea, guanidine thiocyanate, NaI, guanidine isothiocyanate, D-/L-arginine, hydrogen perchlorate or perchlorate salt of Li⁺, Na⁺, K⁺, or a combination thereof. In a particular embodiment, the chaotropic compound is GnCl. The nucleic acids can be contacted with the matrix in the presence of, for example, about 3.5 M to about 6 M of GnCl, or about 4 M to about 5 M of GnCl, or about 4.4 M of GnCl.

(97) In some embodiments, the nucleic acids are contacted with the matrix in the presence of a chelating compound selected from EDTA, EGTA, citric acid, TPEN, 2,2'-Bipyridyl, DFOM, tartaric acid, or a combination thereof. In a particular embodiment, the chelating compound is EDTA.

(98) In some embodiments, the nucleic acids are contacted with the matrix in the presence of a detergent selected from Triton X-100, Tween 20, N-lauroyl sarcosine, SDS, dodecyltrimethylphosphine oxide, sorbitan monopalmitate, decylhexaglycol, 4-nonylphenyl-polyethylene glycol, or a combination thereof. In a particular embodiment, the detergent is Triton X-100. The nucleic acids can be contacted with the matrix in the presence of, for example, about 3% to about 6% of Triton X-100, or about 4% to about 5% of Triton X-100, or about 4.5% of Triton X-100.

(99) In some embodiments, the nucleic acids comprise maternal nucleic acids or fetal nucleic acids.

In some embodiments, the nucleic acids are cell free nucleic acids or circulating tumor nucleic acids. In some embodiments, the cell free nucleic acids are obtained from a sample of a maternal blood, plasma, or serum. In some embodiments, the cell free nucleic acids comprise, for example, cell free fetal DNA and cell free maternal DNA.

(100) The nucleic acids can be, for example, about 50 to about 1200 base pairs in length, or about 70 to about 500 base pairs in length, or about 100 to about 200 base pairs in length, or about 130 to about 170 base pairs in length. In one embodiment, the nucleic acids comprise DNAs. In another embodiment, the nucleic acids comprise RNAs.

(101) In some embodiments, the matrix comprises siliceous materials, silica gel, glass, glass fiber, zeolite, aluminum oxide, titanium dioxide, zirconium dioxide, kaolin, gelatinous silica, magnetic particles, ceramics, polymeric supporting materials, and or a combination thereof. In a particular embodiment, the matrix comprises glass fiber.

(102) In some embodiments, the method further comprises incubating a biological sample comprising the nucleic acids with a protease such as proteinase K, prior to contacting the nucleic acids with the matrix. The biological sample can be, for example, a sample of a maternal blood, plasma, or serum.

(103) In some embodiments, the method further comprises washing the matrix with at least one washing buffer to remove impurities. In some embodiments, the method further comprises drying the matrix. In some embodiments, the method further comprises eluting the nucleic acids from the matrix with an elution buffer.

(104) In some embodiments, the contacting step binds at least 40%, at least 50%, or at least 60%, or at least 70%, or at least 80%, or at least 90%, of nucleic acids having a length of about 72 bp that are present in the composition to the matrix. In some embodiments, the contacting step binds at least 40%, at least 50%, or at least 60%, or at least 70%, or at least 80%, or at least 90%, of nucleic acids having a length of about 118 bp that are present in the composition to the matrix. In some embodiments, the contacting step binds at least 40%, at least 50%, or at least 60%, or at least 70%, or at least 80%, or at least 90%, of nucleic acids having a length of about 194 bp that are present in the composition to the matrix. In some embodiments, the contacting step binds at least 30%, at least 40%, or at least 50%, or at least 60%, of nucleic acids having a length of about 50 bp that are present in the composition to the matrix.

(105) Additional embodiments of described herein relate to a kit for isolating nucleic acids from a biological sample, comprising a binding buffer, wherein the binding buffer comprises a chaotropic compound and a solvent, wherein the solvent comprises an aprotic solvent such as a nitrile compound, tetrahydrofuran, or a combination thereof.

(106) In some embodiments, the binding buffer comprises a nitrile compound selected from ACN, PCN, BCN, IBCN, or a combination thereof. In a particular embodiment, the binding buffer comprises ACN. The binding buffer can comprise, for example, about 15% to about 35% of ACN, or about 20% to about 30% of ACN, or about 25% of ACN.

(107) In some embodiments, the binding buffer comprises less than 5% of alcohol, or less than 2% of alcohol, or less than 1% of alcohol, or less than 0.1% of alcohol, or comprises no alcohol. In some embodiments, the binding buffer comprises less than 5% of propanol, or less than 2% of propanol, or less than 1% of propanol, or less than 0.1% of propanol, or comprises no propanol such as isopropanol. In some embodiments, the binding buffer comprises less than 5% of non-water protic solvents, or less than 2% of non-water protic solvents, or less than 1% of non-water protic solvents, or less than 0.1% of non-water protic solvents, or comprises no non-water protic solvents. The pH of the binding buffer can be, for example, about 4 to about 10, or about 4 to about 5, or about 5 to about 6, or about 6 to about 7, or about 7 to about 8, or about 8 to about 9, or about 9 to about 10, or about 4 to about 8, or about 4.5 to about 6, or about 4.9 to about 5.1.

(108) In some embodiments, the binding buffer comprises a chaotropic compound selected from GnCl , urea, thiourea, guanidine thiocyanate, NaI , guanidine isothiocyanate, D-/L-arginine, a

perchlorate or perchlorate salt of Li⁺, Na⁺, K⁺, or a combination thereof. In a particular embodiment, the binding buffer comprises GnCl. The binding buffer can comprise, for example, about 5 M to about 8 M of GnCl, or about 5.6 M to about 7.2 M of GnCl, or about 6 M of GnCl. (109) In some embodiments, the binding buffer comprises a chelating compound selected from EDTA, EGTA, citric acid, TPEN, 2,2'-Bipyridyl, DFOM, tartaric acid, or a combination thereof. In a particular embodiment, the binding buffer comprises EDTA.

(110) In some embodiments, the binding buffer comprises a detergent selected from Triton X-100, Tween 20, N-lauroyl sarcosine, SDS, dodecyldimethylphosphine oxide, sorbitan monopalmitate, decylhexaglycol, 4-nonylphenyl-polyethylene glycol, or a combination thereof. In a particular embodiment, the binding buffer comprises Triton X-100. The binding buffer can comprise, for example, about 1% to about 6% of Triton X-100, or about 2% to about 4% of Triton X-100, or about 3% of Triton X-100.

(111) In some embodiments, the kit further comprises a digestion buffer comprising a protease such as proteinase K for digesting a biological sample. In some embodiments, the kit further comprises a washing buffer for washing the matrix to remove impurities. In some embodiments, the kit further comprises an elution buffer for eluting the nucleic acids from the matrix.

(112) The binding buffer described herein can be used in a process for binding nucleic acids to a matrix, wherein the binding buffer is mixed with a biological sample (e.g., blood, plasma, or serum) that has been pre-treated with a digestion buffer comprising a protease such as proteinase K.

WORKING EXAMPLES

Example 1—Solvent System for Isolating cfDNA

(113) 1.1—Plasma Separation from Whole Blood

(114) For each pair of blood collection tubes (BCT's) label one 15 mL conical tube and one 50 mL conical tube with the corresponding sample ID. Centrifuge BCTs at 2,000 rcf for 20 minutes at 22° C. to separate plasma from cells. Recover plasma from each BCT tube, without disturbing the pelleted cell layer, with a 10 mL serological pipette and transferred to a single 15 mL conical tube and remove remaining cell debris with a second 30 minute clarifying spin at 3,220 rcf at 22° C. Transfer the clarified plasma to 50 mL conical tubes avoiding pelleted material. Record volume and hemolysis grade for each plasma (i.e., yellow=None, pink/orange=Moderate, and red/dark red=Severe). Low volume (<6 mL) and severely hemolyzed plasma samples should not be processed. Begin the extraction process of plasma samples immediately or store frozen at -80° C.

(115) Reagents:

(116) TABLE-US-00001 Formula Reagent Proteolysis Triton X100 (Triton) Buffer Guanidine chloride (GnCl) Tris chloride (Tris-Cl) Ethlenediaminetetraacetic acid solution (EDTA) Proteinase K Proteinase K from Tritirachium album Binding Guanidine chloride (GnCl) Buffer Ethlenediaminetetraacetic acid solution (EDTA) Acetonitrile (ACN), Ethanenitrile, Ethyl nitrile, Cyanomethane, Methyl cyanide Triton X100 (Triton) Wash Ethanol Buffer 1 N-Lauroylsarcosine (NLS) Tris chloride (Tris-Cl) Ethlenediaminetetraacetic acid solution (EDTA) Distilled Water Wash Ethanol Buffer 2 Tris chloride Buffer Solution (Tris-Cl) Ethlenediaminetetraacetic acid solution (EDTA) Distilled Water Elution 10 mM Tris, 0.1 mM EDTA (pH 8) Buffer

1.2—Plasma Proteolysis/Establishing Proteinase K Digestion Conditions

(117) Adjust the volume of fresh or thawed frozen plasma samples to 10 mL with 1×PBS and process immediately. Samples may be held at room temperature for up to 1 hour at room temperature or placed at 4° C. for wait times <12 hours. Prepare a 20 mg/mL Proteinase K solution less than 30 minutes prior to use. Reconstitute each 100 mg lyophilized vial of Proteinase K (PK) by adding 5 mL dH.sub.2O followed by pipetting up and down at least 5× to completely wet the dried protein pellet. Close each PK vial and invert 10× to thoroughly dissolve the protease pellet and place on ice for at least 5 minutes to ensure complete dissolution. Gently flick or shake contents to the bottom of each vial and for consistency pool multiple vials to homogenize and place immediately on ice.

(118) Initiate plasma proteolysis by adding 400 uL freshly prepared Proteinase K solution to each 10 mL plasma sample, cap and inverted each tube 5× to thoroughly mix. Place tubes back into racks at room temperature and proceed until PK has been added to all samples. Without delay, open caps and add 5 mL of PK Proteinase Buffer to each sample one at a time, quickly recap and mix by vortex at high speed for 5 seconds. Arrange samples in racks and submerge in a 42° C. water bath until the water level reaches at least three quarter height of the digestion mix and incubate for 45 minutes. Once the Proteinase K digestion process is complete, immediately move to the next step—Establishing the Nucleic Acid Binding State.

(119) TABLE-US-00002 TABLE 1 Composition and Ranges for Enzymatic Plasma Proteolysis by Proteinase K Reagents Range Plasma (Sample) 61.7-68.2% Tris-Cl 10-15 mM EDTA 2.5-10 mM Guanidine chloride 1.8-2.2 M Triton X100 5%-8% Proteinase K 0.4-0.6 mg/mL

1.3—Establishing the Nucleic Acid (NA) Binding State

(120) Remove racks from the water and blot dry. If samples are to receive quantification targets, add the requisite amount of spike material to test samples, recap, and mix thoroughly. Uncap tubes and add Binding Buffer to each, recap, invert 10× to mix contents, and place back into the water bath at 42° C. for 10 minutes. This step completes the lysis process and sets up a chemical environment which favors binding of nucleic acids to solid phase glass fiber or silica supports. Remove the plasma lysates from the water bath, blot dry, and cool at room temperature (18-22° C.) for 10 minutes in preparation for Nucleic Acid Capture by Glass Fiber Vacuum Filtration

(121) TABLE-US-00003 TABLE 3 Composition and Ranges for the Nucleic Acid Binding State Reagents Range* Plasma (Sample)** 27.5-30.5% Tris-Cl 1.5-2.2 mM Proteinase K (Inactivated) — Triton X100 4%-5% EDTA 3-5 mM Guanidine chloride 4.2-4.5 M Acetonitrile 13-18%

*Ranges listed are working ranges expected to give high level recovery of short cfDNA fragments.

**Reagents listed in “italics” are carried over from proteolysis and are not present in Binding Buffer.

1.4—Nucleic Acid Capture by Glass Fiber Vacuum Filtration

(122) Prepare glass fiber spin columns for filtration by labeling and fitting a disposable plastic vacuum connector to the exit port. The connectors prevent spin column contamination from the vacuum manifold. Install spin columns on the vacuum manifold and check that all connections are secure. Plug any unused vacuum ports and connect vacuum lines to the manifold and keep the pressure at zero mBar. Wet each column by carefully pipetting 500 µL of Spin Column Conditioning Solution onto the center of each membrane without directly contacting the membrane with the pipette tip. Engage the vacuum briefly to initiate a slow flow of the conditioning solution through the columns. Once complete, interrupt the vacuum. Attach a 45 mL Column Extender to each column and check to make sure the connections are snug. Initiate NA binding by carefully pouring plasma lysates in the nucleic acid binding state into reservoir extenders and initiate filtration by bringing the vacuum to -600 to -800 mBar. Filtration times may vary from sample to sample, but should complete within 45 minutes, and not typically less than 10 minutes. Wash both columns as described below and elute sequentially with 55 uL elution buffer passed over both columns (1.sup.st Pass and 2.sup.nd Pass), recovering the eluate in a separate tube for each binding matrix.

(123) 1.5—Sequential Wash Steps, Residual Wash Removal and Drying

(124) Once filtration of all plasma binding lysates is complete, remove the reservoir extender from each spin column, and add 850 uL of Wash Buffer 1 to each spin column. Release the vacuum, bring the pressure to 0 mBar, and add 825 µL of Wash Buffer 2 and reengage the vacuum to draw wash buffer through column. Turn off the vacuum and allow the pressure to reach 0 mbar and add 825 uL of 100% ethanol resume filtration under a vacuum of -600 mBar. Once filtration is complete, allow columns to dry under vacuum for 1 minute, then deduce the vacuum pressure to 0 mBar and close the lid of each spin column. Take each column off the vacuum manifold, remove the disposable vacuum connectors, and place each into a clean 2.0 mL collection tube. Load into a

microcentrifuge and spin at 14,000 rpm for 3 minutes to dry residual EtOH. Preheat Elution Buffer to 56° C. prior to elution. Transfer each spin column to a 1.5 mL pre-labeled LoBind microcentrifuge tube.

(125) 1.6—NA Elution from Glass Fiber Spin Columns

(126) Add 50 μ L of pre-heated Elution Buffer to the center of each filter without touching the filter membrane with the pipette tip. Close spin column lids and incubate at room temperature (18° C. to 22° C.) for 7-10 minutes. Elute cfDNA by centrifugation at 14,000 rpm for 1 minute. Recovered cfDNA can be taken directly into NGS library preparation or stored at -20° C. for future analysis.

(127) 1.7—Comparative Testing

(128) As shown in FIG. 1, when IPA was used as solvent, recovery of short dsDNA increased as the IPA concentration was raised and the GnCl concentration fell due to volume displaced by the added solvent.

(129) As shown in FIG. 2, when ACN was used as solvent, recovery of short dsDNA increased as the ACN concentration was raised and the GnCl concentration fell due to volume displaced by the added solvent.

(130) As shown in FIG. 3, recovery of short dsDNA increased as the concentration of GnCl was increased relative to a constant amount of ACN (8.3% ACN in the NA binding state).

(131) As shown in FIG. 4, recovery of short dsDNA decreased as the concentration of GnCl or percentage of ACN was reduced. An increase of either ACN or GnCl can compensate for the insufficiency of the other. For instance, recovery of the 72 bp and 118 bp fragment improved when ACN was held constant at 11% and GnCl was increased from 4.19 M to 4.47 M, and also when GnCl was held constant at 4.19 M and ACN was increased from 11% to 16.6% in the nucleic acid binding state.

(132) As shown in FIG. 5, increased concentration of Triton in the binding condition yielded an increase in spike recovery of the 72 bp exogenous DNA target, under two different test conditions (13.8% ACN and 4.24 M GnCl , or 16.6% ACN and 4 M GnCl).

(133) As shown in FIG. 6, increased concentration of Triton in the binding condition yielded an increase in recovery of small DNA fragments from 2 single donor plasma samples under two different test conditions (13.8% ACN and 4.24 M GnCl , or 16.6% ACN and 4 M GnCl).

(134) As shown in FIG. 7, various water soluble solvents other than ACN were added to create binding conditions that could promote increased recovery of the 72 bp DNA fragment.

(135) As shown in FIG. 9, fetal fraction estimates for matched pair pregnancy samples were significantly increased when cfDNA was isolated when ACN served as a co-solvent in comparison to IPA in the nucleic acid binding state.

(136) As shown in FIG. 10, filtration times through glass fiber filters were significantly shorter for match paired plasma samples when the nucleic acid binding state was established with ACN as a co-solvent compared to IPA.

Example 2—High-Precision Size Selection & Purification of cfDNA

(137) FIG. 14 shows an exemplary workflow of dual filtration size selection using low and high stringency binding buffers (LSBB, HSBB). The lysis portion consists of a 10 mL plasma from whole blood in a 50 mL conical tube, to which 400 μ L of 20 mg/mL Proteinase K 6.5 mL of proteinase K digestion buffer (PKDB) was added. The solution was vortexed and incubated at 42° C. for 45 minutes to digest plasma proteins. Once lysis was complete, a low stringency binding buffer (LSBB) was added to the solution, creating the ideal environment for binding of larger DNA fragments to the glass fiber membrane. The LS lysate was cooled to room temperature and passed through an immobilized silica membrane, and the first pass filtrate was collected into a 50 mL conical tube by use of a device known as a VacCap. The first pass flow-through was supplemented with a high stringency binding buffer (HSBB) that established a binding condition for membrane capture of the remaining DNA species. The spent flow-through was treated as waste. Wash and elution steps followed and both captured fraction were eluted and saved.

(138) An proof-of concept experiment was performed as shown in FIG. 20 (right panel) to demonstrate that the sequential two-step capture can be successfully applied for isolation of cfDNA. Two plasma samples of the same origin were utilized. The first was processed with the single-filtration comparative workflow shown in FIG. 13 (i.e., total, single stage DNA capture and elution). The pH of the binding lysate in the comparative workflow was 5.78.

(139) The second plasma sample was treated to the two-step binding protocol shown in FIG. 14 in which the pH of the plasma lysate was adjusted to pH 7.17 prior to first vacuum filtration step. The flow-through was recovered pH adjusted to 5.32 with malic acid and the lysate subjected to a second filtration step. The objective was to reduce the pH to below the control sample (5.78) to ensure capture of the remaining DNA spike targets.

(140) As shown in FIG. 15, DNA fragments were differently captured based on the pH of the lysate prior to immobile phase capture, which proves that the dual filtration method can be applied for partitioning and thus enriching cfDNA fragments directly from plasma during the nucleic acid purification step. The electropherogram of FIG. 16 shows plasma cfDNA and co-purified defined fragments and demonstrates that the major cfDNA peak around 160 bp in size is bracketed by the spike targets that were quantitatively tracked by quantitative polymerase chain reaction (qPCR). All the peaks observed in the electropherogram are ladder fragments that make up the tracked spike fragments.

(141) FIG. 17 shows data from an experiment that tested low stringency binding conditions by uniform titration of components to progressively increase the stringency of the first pass. It was found that incremental increases altered in a predictable way the selective retention of the 118 and 194 bp spike fragments.

(142) Seeing that pH had an effect on size selection, the observations from the experiment of FIG. 15 and the experiment of FIG. 17 were combined to improve the ability to resolved DNA fragments directly from plasma. FIG. 18 shows that by increasing the pH of the low stringency binding condition, in ever increasing GnCl, Triton, and ACN concentration, we can adjust size selection to increase separation between small fragment capture compared to larger DNA fragments.

(143) There are two experiments shown in FIG. 19. The top panel of FIG. 19 demonstrates the difference between low stringency binding without ACN or with ACN. Rows 2 has recorded the first Pass capture and row 3 has recorded the second Pass percent capture results without ACN in Pass 1. Likewise, Rows 4 and 5 show first and second Pass percent capture with ACN added. Higher efficiency of DNA capture was achieved with ACN present in the LS lysate, suggesting benefit for minimizing small fragment capture as evidenced by less 72 bp fragment capture (compare rows 3 and 5).

(144) The second experiment looked into whether a higher pH of the first pass lysate would be even more beneficial. As shown in the bottom panel of FIG. 19, the higher pH condition pH 10 binding buffer did not increase the binding of larger DNA fragments (compare rows 5 and 3, lower table). It was concluded that the lysate pH obtained by adding 4 mL of pH 9 binding buffer was more preferably than that obtained by adding 4 mL of pH 10 binding buffer.

(145) Results from the two-step filtration experiments suggested that it is possible to preferentially enrich for fetal cfDNA directly from maternal plasma, and this was tested on twelve 20 mL plasma samples from pregnant mothers. FIG. 20 shows how, for comparison, each plasma sample was divided into two 10 mL aliquots. One aliquot was treated to the standard one-step purification method outlined in FIG. 13. The matched sample was purified by the two-step method shown in FIG. 14.

(146) FIG. 20 (left panel) shows a heat map of the percentage of spiked fragments captured by the first and second pass filtration step under low and high stringency conditions, respectively. The second pass filter, labeled second Elution is presented in the top half of the table. The size of fragments tracked is listed above the last 8 columns. The Reaction header in column 1 indicate the numbering of the paired sample, pair 1, 2, 3, and so on. This experiment demonstrated the

astounding degree to which DNA lost (not captured) in the low stringency first pass, but is subsequently captured entirely by high stringency second pass. This demonstrates that a high degree of discrimination can be achieved between fragments differing by as little as 80 bp, with near complete recovery of all DNA fragments in either the first or second pass elutions. This means a powerful new method for sized selection at purification (SSAP) that is highly efficient and does not result in the loss of any of the original nucleic acids in the starting sample, in this case plasma. The percentage recovery data are summarized graphically in FIG. 21.

(147) FIG. 22 shows results following analysis of isolated cfDNAs, Control vs Test (first pass and second pass). The recovered cfDNA population for each case were processed through Natera's Panorama v3 pipeline. The results demonstrate that the detected child fraction estimate (% CFE) column 8, for all second pass samples were higher than the matched control. Reciprocally, all first pass samples returned lower % CFE than the matched control. This latter result fits with the prediction that omitting smaller cfDNA fragments from a maternal plasma sample would reduce the fraction of child cfDNAs which have been demonstrated to be smaller on average (144 bp vs 160 bp) compared to maternal cfDNA. This method demonstrates the clear ability to achieve a relative increase in fetal fraction in this simple, elegant 2-step capture that ranged from 4% to several folds over control. The CFE data in FIG. 22 are summarized graphically in FIG. 23.

(148) FIG. 23 is a bar graph of the pregnancy plasma samples described in FIG. 22. It demonstrates that for all 12 cases, an increase in CFE was observed in the second pass while a reduction in % CFE was observed in the first pass. This was predicted. In addition, the lower capture of the 118 bp SQA spike fragment can be correlated to increased fetal fraction in the second pass. This fits with the best evidence that suggests fetal cfDNA fragments are smaller than maternal cfDNA.

(149) FIG. 24 shows hetrate plots for maternal plasma sample #3. The Control (top) isolate, and matched Test isolates from second pass (middle) and first pass (bottom) are depicted in the top, middle and bottom panels. There are two key takeaways from these hetrate plots. The first is that hetrate plots can be used to determine fetal fraction. Every dot represents one of the Panorama v3~14K SNPs. The middle plot (second Pass) cfDNA has two extra horizontal bands near the top and bottom axes. These two bands are the fetal cfDNA bands. The higher the child fraction, the closer to the center these two bands are located. These bands are present in the top and (Control) and bottom (first Pass) plots, but because their respective child fractions are much lower, they remain nearer the upper and lower axes compared to the center (second Pass) plot. Second, shifts in the plot demonstrate ploidy type. An extra chromosome causes very apparent shift in hetrate plots. Such shifts were not observed here because this sample represents a normal (euploid) condition. One of the considerations with the dual filtration method is that large child fraction increases can result in fewer molecules of DNA entering the library construction process. This is reflected in Natera's "noise parameter" (FIG. 22, column 9). However, despite the low noise parameter metric, all 13,392 SNPs of Panorama v3 obtained good coverage and the number of reads received per SNP was just as uniform as its control sequencing sample.

(150) FIG. 24 demonstrated that pregnancy plasma #3, which observed the highest increase in child fraction, rising from 6.31% for the control isolation procedure, to 18.14% for the 2-step size selection at purification (SSAP) procedure. Even though this sample would have failed the noise parameter metric, potentially due to input of unique DNA molecules from the mother, a very clean hetrate plot was obtained. It means this second pass sample was not "noisy" as would be expected of samples with such low noise parameter metrics. With apparently much higher fetal derived DNA in this isolate, and the excessive maternal cfDNA and gDNA, making a ploidy type call with this hetrate plot can be accomplished with greater accuracy.

(151) FIG. 25 is an electropherogram of the library produced from sample #1's control, second pass, and first pass samples overlapped. These libraries, due to the addition of ligated amplification sites, are larger in fragment sizes than cfDNA. However, it was demonstrated that by selecting for different sizes of DNA, different peaks can be observed per sample. The second pass sample, which

selected for smaller DNA, produced smaller libraries with additional shoulder peaks around ~220 to 240 bp. Meanwhile, the first pass sample, which emphasized capture of larger cfDNA, produced larger libraries with additional larger shoulder peaks around ~270 bp. The child fractions of 11.82% for control, 13.63% for second pass, and 10.32% for first pass, supporting that the selection of differently sized cfDNA can attribute to differences in fetal fraction.

Example 3—No Salt Increase—Size Exclusion Methods

(152) This example demonstrated the size selection process, wherein differential binding is achieved by altering the chemical environment of the plasma lysate (the mobile phase) prior to contacting a binding substrate (the immobile phase) such as glass fiber filters, silica membranes, or silica beads.

(153) The foregoing examples 1 and 2 demonstrated the principle of simultaneous purification size selection in which, coincident with purification, double stranded cell-free DNA (ds-cfDNA) fragments are isolated directly from crude plasma lysates and partitioned into distinct fractions containing subpopulation that differ in their average size (length in base pairs (bp)). As demonstrated in preceding examples 1 and 2, the size cutoff, defined as the approximate length (bp) above which the affinity of double stranded DNA (dsDNA) for the immobile phase is high enough for binding to occur, can be varied substantially. Without being bound by theory, it is an hypothesis herein that there are a multitude of physical-chemical properties that affect the relative affinity of double stranded DNA for glass fiber filters (borosilicate glass) or silica membranes, and no single binding theory can describe the myriad interactions that balance the binding equilibriums that dictate length dependent binding of dsDNA fragments.

(154) Herein, a SSAP system was developed by empirically determining which components in the low stringency and high stringency binding buffers used had the greatest influence over length dependent binding of dsDNA fragments to glass fiber filters. Among the most notable effectors of the binding response were pH, chaotropic salt concentration, solvent polarity, and detergent type and concentration. Response profiling indicated that pH, solvent type and solvent concentration were the most controllable effectors of length dependent glass filter binding of dsDNA in a plasma lysate background. It was found that controlling the size cutoff in the low stringency binding conditions was the most important factor for size selection as it determined the extent to which fragments of a certain size are included or excluded from a given fraction. It was also found that solvent and pH brought about both gradual and broad ranging binding responses, and by modulating either or both components, the cutoff size could be increased or decreased to some degree, and the sharpness of the cutoff could also be controlled.

(155) These observations gave rise to a new process in which pH and solvent were adjusted to achieve variable low stringency and high stringency binding conditions that favored or disfavored the binding of dsDNA fragments above or below a certain length (bp). This was accomplished while keeping the concentration of chaotropic salts in the low and high stringency binding conditions constant, and thus, the disclosure herein established a variation of the SSAP system that is referenced in the following as No Salt Increase—Size Selection At Purification or NSI-SSAP (FIGS. 26-30) when applied to plasma samples, and No Salt Increase—Library Size Selection or NSI-LSS (FIGS. 31-36) when applied to cfDNA preserved as adapter ligated libraries.

Methods and Materials

(156) An example of the NSI-SSAP workflow employing dual-filtration size selection under low and high stringency binding conditions in which salt concentration was not increased is illustrated in FIG. 26. The concentration of chaotropic salt in the low stringency binding condition begun at >30% and remained unchanged, or decreased slightly, when the subsequent high stringency binding condition was formed. To begin, an initial lysis portion was established by combining 10 mL plasma from whole blood with 200 μ L of 20 mg/mL Proteinase K and 11.7 mL of low stringency binding buffer (LSBB) in a 50 mL conical tube. The mixture was incubated at 42° C. for 45 minutes to digest plasma proteins and the lysate was cooled to ambient temperature (15-25° C.) and

filtered through a glass fiber or silica membrane. The filtrate was collected with a device like the one shown and saved. The first pass filtrate was mixed with 22 mL high stringency binding buffer (HSBB) to establish the high stringency binding condition, and filtered again through a glass fiber filter or silica membrane to capture the remaining DNA fragments. Each filter was washed and eluted to recover the 1st Pass, Large cfDNA Fraction and 2nd Pass, Small cfDNA Fraction. The filtrate from the 2.sup.nd column may be either be discarded or saved for further processing.

(157) For the NSI-SSAP method, the low stringency binding buffer (LSBB) composition may comprise between about 0 to about 20% Acetonitrile, about 0 to about 15% Tween 20, about 6 to about 7.8 molar Guanidine chloride, about 10 to about 50 millimolar Tris(hydroxymethyl) aminomethane (Tris, free base), about 0 to about 2 mM Ethylenediaminetetracetic acid (EDTA, free acid), and have a pH of about 7.0 to about 10. Correspondingly, the high stringency binding buffer (HSBB) composition may comprise between about 10 to about 40% Acetonitrile, about 0 to about 10% Tween 20, about 0 to about 2% Ethanol, about 3.2-about 3.6 molar Guanidine chloride, about 10 to about 70 millimolar 2-(N-morpholino)ethanesulfonic acid (MES, free acid), about 0 to about 2 mM Ethylenediaminetetracetic acid (EDTA, free acid), and have a pH of about 4.05 to about 6.5.

(158) In the NSI-SSAP method, the low stringency binding condition also establishes conditions for protease digestion and is prepared by combining 10 mL plasma, serum, urine, or other cell free biological fluid, with about 50 to about 300 uL proteinase K (20 mg/mL) and about 11.7 to about 15.7 mL low stringency binding buffer (LSBB). In one preferred embodiment the final concentrations are listed in FIG. 27 (upper table), and concentrations of various components can range accordingly: plasma (about 38.9% to about 45.66%), Acetonitrile (about 0% to about 12.1%), Tris base (about 5.43 mM to about 20 mM), Guanidine chloride (about 3.2055 M to about 4.76 M), Tween 20 (about 0% to about 9%), EDTA (about 0% to about 1.2%) and a pH of about 7 to about 9. Following an incubation time of 30 minutes to 2 hours at temperature between 20° C. and 45° C., the lysate was filtered through a glass fiber filter or silica membrane and the filtrate collected and saved to purify the remainder of cfDNA. During the filtration step a variable amount cfDNA fragments will bind to the filter membrane depending on the length (bp) of the dsDNA fragment. The saved filtrate from the 1.sup.st filtration is then mixed with additional reagents to convert it to a higher stringency binding condition. In the examples given, the high stringency binding condition is formed by mixing the saved ~21.9 to ~25.9 mL of filtrate with ~22 to ~26 mL of HSBB. In one preferred embodiment, final concentrations in the high stringency case have been listed (FIG. 27, lower table), and concentrations of key components may vary: plasma (about 19.27% to about 22.78%), Acetonitrile (about 4.59% to about 26.09%), Guanidine chloride (about 3.2055 M to about 4.76 M) with concentrations holding constant at low and high stringency, Tween 20 (about 0 to about 9.55%), Tris-base (about 1.22 mM to about 5.33 mM), MES (about 4.59 mM to about 38 mM), Ethanol (about 0 to about 1%), EDTA (about 0 to about 0.61 mM), and a pH of about 4.05 to about 5.5. This lysate was filtered through a glass fiber or silica membrane to capture the cfDNA therein. The NSI-SSAP method did not lead to the loss of any cfDNA from the sample during purification, but instead partitions all recoverable cfDNA into either the large or small cfDNA fractions, eluted from the 1.sup.st or 2.sup.nd glass fiber membranes following standard wash, dry and elution steps.

(159) The compositions of the low and high stringency binding conditions established in the methods depicted in FIG. 26 and FIG. 31 are shown in FIG. 27. The experimental results deriving from these conditions are presented in FIGS. 28 through 35. The components and concentrations established with LSBB when added to plasma or library products in the proportions described in FIG. 26 and FIG. 31 are shown in FIG. 27 in the upper table, whereas. the lower table lists components and concentrations established following addition of HSBB to lysates recovered following 1st Pass vacuum or 1st Spin Column filtration.

Results

(160) Evidence that modulation of solvent type and concentration can strongly affect length-

dependent affinity of dsDNA fragments in crude lysates for glass or silica substrates is presented most clearly in FIGS. 4, and 7. These experiments demonstrated that the percentage of 72 bp and 118 bp fragments recovered depended on the mobile phase concentration of the solvent acetonitrile, as well as propionitrile, butyronitrile, and isobutyronitrile.

(161) The effect of pH on the length dependency of glass fiber or silica membrane binding was shown in FIG. 28, where the percentage of 310 bp fragment (checkerboard pattern) decreased with increasing pH. Such data indicated that multiple parameters can affect the affinity of dsDNA for finely spun borosilicate glass fibers or silica membranes, and that DNA length itself has a profound effect on the strength of the interaction.

(162) The % recovery data obtained for fragments of DNA size 72, 118, 194, 310, and 1078 base pairs that were spiked into plasma lysates and eluted from the 1st pass columns (i.e., the large DNA fragments) is shown in FIG. 28. In the series presented, the spiked plasma lysates were titrated with a buffered Tris solutions to obtain pH values between 6.6 and 7.15 within the low stringency binding condition. As the pH of the low stringency binding condition increases, the affinity of the shorter DNA fragments for the glass fiber filter or silica membrane decreases. Reciprocally, as pH decreases the affinity of shorter DNA fragments for the glass fiber filter or silica membrane, increases. This differential affinity is achieved following the addition of high stringency binding buffer (HSBB) without changing the concentration of the chaotropic salt.

Example 3.1—Using Size Selections at Purification (NSI-SSAP) to Enrich for the Fetal Cell-Free DNA from Maternal Plasma Samples

(163) Next, we evaluated if the size exclusion methods developed herein can enrich for fetal cell-free DNA from maternal plasma samples. The average length of cell-free DNA fragments originating from the child and present in the maternal circulation are shorter, ~143 bp, compared to the average cfDNA of the mother, ~166 bp as previously reported (Chan K C A, Zhang J, Hui A B, et al. (2004) Size distributions of maternal and fetal DNA in maternal plasma, *Clin Chem.*

50(1):88-92, and Fan H C, Blumenfeld Y J, Chitkara U, Hudgins L, Quake S R (2010) Analysis of the size distributions of fetal and maternal cell-free DNA by paired-end sequencing, *Clin Chem.* 56(8):1279-1286). The percentage of fetal DNA present in maternal blood is on average only 10%, but frequently less than 5% in plasma samples, particularly in plasma collected in the 1st trimester of pregnancy.

(164) As with SSAP, the length discrepancy should make it possible to increase the apparent child fraction by enriching for cfDNA fragments at the time of purification. NSI-SSAP was used to simultaneously purify and size select cfDNA from the blood plasma from 10 pregnant women. The cfDNA was preserved and analyzed in the single nucleotide polymorphism (SNP) based non-invasive prenatal test (NIPT) Panorama (Samango-Sprouse C, Banjevic M, Ryan A, et al. (2013) SNP-based non-invasive prenatal testing detects sex chromosome aneuploidies with high accuracy. *Prenatal Diagnostics* 33:643-9, and Hall M P, Hill M, Zimmermann, P B, et al (2014) Non-invasive prenatal detection of trisomy 13 using a single nucleotide polymorphism- and informatics-based approach. *PLoS One* 9:e96677). The Panorama™ assay may be used to calculate the proportion of fetal to maternal SNP's, accurately reported as the percent child fraction estimate (% CFE). If shorter cfDNA fragments are indeed enriched in the small cfDNA fraction, the proportion of child SNP's should be higher than the control from a non-size selected cfDNA fraction. Reciprocally, the % CFE in the large fraction should be reduced compared to control.

(165) To test if NSI-SSAP method could enrich for fetal cfDNA, 10 mL of pregnancy plasma (10 unique cases), was processed by the NSI-SSAP method. Proteolysis was initiated with the additions of 200 μ L of 20 mg/mL Proteinase K and 11.7 mL of low stringency binding buffer (LSBB) added to each 10 mL plasma. The addition of these two components established the low stringency binding condition, which caused larger DNA fragments to bind glass fiber filter/silica membranes, in conditions that, at the same time, disfavored binding of smaller DNA fragments also present in the lysate. In this example, the majority of DNA spike fragments >194 bp bound with high

efficiency to the 1st pass filter. Next, high stringency binding buffer (HSBB) was added to the 1.sup.st Pass filtrate to establish the high stringency binding condition where now the smaller DNA fragments that did not bind to the 1st filter can be efficiently captured by the 2.sup.nd pass glass fiber filter/silica membrane. It was found that 72, 118, and some 194 bp, were detected in high percentage in the 2nd pass eluates recovered for each case. The percentage of spike targets recovered in the large fraction (1st column elution) and small fraction (2nd column elution) are shown in FIG. 29. The size selection obtained with this method showed exclusion of large DNA fragments in the small fraction, which was directly correlated with increased child fraction estimates (% CFE), listed below each case as shown in FIG. 29. Thus, it was found that the small fraction was enriched for fragments of cfDNA of fetal origin.

(166) In summary, the data shown in FIG. 29 demonstrated that the NSI-SSAP enriched for cfDNA Small Fractions (2.sup.nd Elution). In particular, NSI-SSAP enriched for the 72 bp and 118 bp spike targets, and this method enriched for fetal SNP's partitioning with cfDNA fragments that were shorter on average. In this example, under the size selection conditions practiced, NSI-SSAP resulted in relative increase in the % CFE in the cfDNA Small Fraction of 87%, which constituted a near doubling of the CFE.

(167) Next a data fit prediction of the % CFE results from the above experiment shown in FIG. 29, were performed and the result is shown in FIG. 30. The plot shown in FIG. 30 compared cfDNA fractions isolated without size selection, the Control (Ctrl), to the 1st Elution (large cfDNA fraction) and the 2nd Elution (small cfDNA fraction), for all cases. This required that two $\times 10$ mL plasma samples be extracted for each case. The cfDNA from one 10 mL aliquot was processed to recover total cfDNA, while the other was processed by the NSI-SSAP method to partition cfDNA into the 1.sup.st Pass filter eluate (large cfDNA fraction) and the 2.sup.nd Pass filter eluate (small cfDNA fraction). Compared to Control, lower % CFE values were obtained for the large cfDNA fractions, and higher % CFE values were observed in the 2.sup.nd Pass (small cfDNA) fractions. Thus, the data analysis shown in FIG. 30 indicated that the small cfDNA fraction is enriched for fetal cfDNA fragments compared to either control or large cfDNA fraction. In other words, the plot shown in FIG. 30 demonstrated that shorter cfDNA fragments are enriched in the small cfDNA fraction, and thus, the proportion of child SNP's (% CFE) is higher in the cfDNA Small Fraction than in the Control, and lower in the cfDNA Large Fraction.

Example 3.2—Using No Salt Increase for Library Size Selection (NSI-LSS) to Enrich for cfDNA Preserved in Molecular Libraries

(168) The experiments shown in FIGS. 29 and 30 demonstrated how the NSI-SSAP method can be aptly applied to enrich shorter cfDNA fragments in a manner that is coincident with purification and directly from plasma. Moreover, the method is easily extendable to dsDNA fragments in other backgrounds. In the following examples the alternate low and high stringency binding strategy will be demonstrated when applied to cfDNA's preserved in molecular libraries.

(169) To produce the library, the ends of each cfDNA, regardless of state, are repaired enzymatically and ligated to DNA adapter molecules to give each cfDNA a common forward and reverse primer binding site at either end. This immortalized the cfDNA and allowed all members of the library to be uniformly amplified many times to increase their numbers without biasing the proportion of any in the population. Amplified libraries from the Panorama NIPT test were taken through a modified protocol at smaller scale. FIG. 31 presented the NSI-LSS adaptation for library size selection in a spin column and centrifuge based method. The method depicted is an exactly scaled version of the plasma NSI-SSAP workflow using identical LSBB and HSBB compositions, so that the concentrations in the low and high stringency binding condition match those detailed in FIG. 27.

(170) The No Salt Increase—Library Size Selection (NSI-LSS) workflow where dual-filtration size selection has been adapted to a spin column format and applied to size select DNA library products into larger and smaller DNA fractions is outlined in FIG. 31. The method utilized the same low and

high stringency binding principles, under which the chaotropic salt concentration in the low stringency binding condition begins at >30% and remains unchanged, or decreased slightly, in the shift to high stringency binding conditions. In the adaptation, 186 μ L of purified or unpurified library sample, typically but not necessarily derived from cfDNA, was combined with 213 μ L of low stringency binding buffer (LSBB) to establish the low stringency binding condition. The mixture was applied to a glass fiber filter or silica membrane spin column and placed into a spin tube as shown. The assembly was centrifuged to force the low stringency binding condition through the glass fiber/silica membrane in the 1.sup.st filtration step, while the low stringency filtrate collected in the lower spin tube below, and was carefully preserved for secondary processing. The 1.sup.st spin column was removed and processed separately to receive a 1 $\times$ wash in Wash Buffer 1, 1 $\times$ wash in EtOH, and eluted with a low ionic strength buffer such as TE (10 mM Tris, 0.1 mM EDTA, pH 8). This partitioned the larger of the library DNA fragments into the “Large Fraction”. The filtrate saved from the 1.sup.st spin was then mixed with 400 μ L high stringency binding buffer (HSBB) to establish a high stringency binding condition. This was applied to a new glass fiber filter/silica membrane spin column and centrifuged to pass the high stringency binding condition across the solid filter membrane to capture the remaining, smaller, library DNA fragments. The 2nd spin column was washed 1 $\times$ with Wash Buffer, 1 $\times$ with EtOH, and eluted in TE to recover the smaller of the library DNA fragments in the “Small Fraction”.

(171) To examine the degree of size separation and obtain an estimate for the size cutoff for the NSI-LSS size selection method practiced in FIG. 31, a dsDNA ladder having fragments in the region of interest (100 bp to 1200 bp) diluted in 10 mM Tris, 0.1 mM EDTA was taken through the spin protocol. Ladder fragments in the original ladder and the large and small fractions were analyzed by capillary electrophoresis (CE) and the traces are presented in FIG. 32.

(172) Capillary electrophoresis (CE) traces were generated on an Agilent BioAnalyzer™ (1K chip) of a broad range (100 bp to 1200 bp) DNA Ladder treated by the NSI-LSS method, as detailed in FIG. 31. Each CE trace included two size markers at 15 and 1500 base pairs, which flank peaks from the DNA ladder. The top trace was of the native ladder showing the starting abundance of each peak. The middle panel showed peaks that were eluted from the 1.sup.st spin column (Large DNA Fraction) in the NSI-LSS workflow. The bottom panel showed peaks eluting from the 2.sup.nd spin column (Small DNA Fraction). The Large DNA Fraction showed a marked reduction in peaks at 200 and 300 bp, and an absence of the 100 bp peak altogether. Conversely, the Small DNA Fraction showed full recovery of all the 100 bp peak and the remaining balance of the 200 and 300 bp peaks not captured in the 1.sup.st pass filtration.

(173) The NSI-LSS size selection method as practiced in FIG. 31 was applied to an amplified library diluted in 10 mM Tris, 0.1 mM EDTA. A sample of the non-size selected starting library and the large and small fractions from the size-selected library were analyzed in parallel by capillary electrophoresis (CE). For overlay clarity, just the traces from the Large and Small fractions are co-presented in FIG. 33. The Large fraction comprised two peaks at 239 bp and 391 bp, consistent with cfDNA molecules of mono-nucleosomal and di-nucleosomal origin. The Small fraction showed the presence of only one peak, representing cfDNA of mononucleosomal origin. The single Small fraction peak was also center shifted by $\sim$ 10 bp compared to the similar peak in the Large fraction, indicating enrichment for library fragments of smaller size.

(174) Capillary electrophoresis (CE) traces were generated on an Agilent BioAnalyzer™ (1K chip), resolving amplified library products size-selected using the NSI-LSS method detailed in FIG. 31. In the Large Fraction Library products eluting from the 1.sup.st spin column there were two peaks because it was comprised of DNA fragments that originated from cell-free (cf) DNAs that were both di-nucleosome and mono-nucleosome in size. The library method added 63 bp to each cfDNA fragment, so that the peaks observed at 239 and 391 bp were in fact quite close to the expected sizes of $\sim$ 230 bp and $\sim$ 380 bp that were generated from mono- and di-nucleosome sized cfDNA fragments. Library products eluting from the 2.sup.nd spin column were devoid of fragments in the

di-nucleosome size range. The peak labeled “Small Fraction Library” migrated with an average length of 229 bp, or ~10 bp shorter than the equivalent library peak eluted from the 1st spin column. This shift portended a slight reduction in the average length of the monosome derived library fragments recovered in the small fraction. The small fraction also appeared to be enriched for even smaller fragments, as the left shoulder of the peak faded slowly off, indicating that there was an abundance of library products originating from the much smaller cfDNA fragments.

(175) To examine the degree of library size selection and obtain an estimate for the size cutoff under the conditions used to generate FIG. 33, the NSI-LSS method was performed on three library samples and the partitioning of 5 spike targets between the Large and Small library fractions was quantitatively tracked by qPCR. The results are presented in FIG. 34 and demonstrated that a majority of fragments larger than 310 bp in size were retained in the Large fraction. Since the average length of fragments in the starting library is 225 bp, the method tested herein likely imposed a very modest degree of size selection among the mono-nucleosomal derived fragments. Still, removing fragments larger in size would be expected to result in a child fraction increased by partitioning proportionally more maternally derived (i.e., on average larger) library cfDNA fragments, particularly those di-nucleosomal in size, into the large fraction.

(176) The percent recovery of spike targets at 72, 118, 194, 310, and 1078 base pairs, that were added to libraries prepared from pregnancy plasma derived cfDNA is shown in FIG. 34. Following library size selection by the NSI-LSS method (FIG. 31), spike recovery of all 5 spike targets was determined by real time quantitative PCR (RT-qPCR). The NSI-LSS method subdivided each library into a Large DNA fraction and a Small DNA fraction. The majority of 310 bp spike targets were detected at a high rate of recovery (66-76%) within the Large Fraction Library eluted from the 1.sup.st spin column. By contrast, the 72, 118, and 194 bp fragments were detected in the Small Fraction Library.

(177) To definitively show that libraries treated to the NSI-LSS method were enriched for shorter cfDNA fragments, the method was applied to 13 libraries generated from the plasma cfDNA from pregnant women early in gestation. Size selected library products in the Large and Small fractions were taken through the Panorama NIPT assay and child fraction estimated obtained shown in FIG. 35.

(178) Next, the NIS-LSS method was applied to amplified libraries generated from 13 pregnancy plasma samples to test the effect on child fraction estimates (CFE). The data are plotted to compare % CFE for control (Ctrl) libraries having no size selection, against the Large Library DNA Fraction (1st spin column elution), and the Small Library DNA Fraction (2nd spin column elution) within the same sample as shown in FIG. 35. A portion of the non-selected library from each pregnancy case served as control. Another portion of the library was treated to NSI-LSS to obtain % CFE for library product returned from the 1st spin and 2nd spin columns. In every case, except case #9, the control and large library fractions (1st spin column eluate) had lower % CFE than the library small library fraction eluted from the 2nd spin column. This demonstrated that smaller DNA fragments that escaped being captured by the 1st spin were subsequently captured by the 2.sup.nd spin column under the higher stringency conditions and, proved to be enriched for smaller fetal DNA fragments.

(179) As expected, the average Small fraction increase in % CFE for the 13 cases presented was modest, with an average relative increase of 8.44%.

Example 3.3—The LibAddition Strategy to Increase the Average Size of the Population of Library Products

(180) It is proposed herein that % CFE may be increased without altering the chemical environment in either the low or high stringency binding conditions by uniformly increasing the overall length of the library itself. Shifting the center of the library size distribution upward, without altering the size cutoff under low stringency, will cause more library fragments to partition into the Large fraction (1.sup.st pass eluate) relative to the Small fraction. This approach is referred

to as the LibAddition strategy and is outlined in FIG. 36.

(181) The LibAddition strategy enables selection of shorter preserved cfDNA fragments without changing the chemistry of the NSI-LSS method. Rather, the library size is increased while the low stringency binding condition remains constant. For example, the NSI-LSS method, as it is applied in FIG. 31 and demonstrated in FIGS. 32 and 34, restricted library products $<\sim 310$ bp in eluates from the 2nd spin column, (the Small Fraction Library). Under these conditions, cfDNA fragments $<\sim 250$ bp in size ($310-63=247$) were enriched.

(182) In the LibAddition strategy a DNA fragment of known length is appended to each library fragment to shift the average size of the population of library products to enrich for cfDNA fragments preserved in the library without altering the chemistry of the method.

(183) In the LibAddition strategy, an additional 25 bp is added to each library product by method known as tailing PCR as shown in FIG. 36. Following one or more rounds of PCR with primers F1/R1-LibAddition, an additional 50 bp is added to each library fragment. Another ~ 50 bp can subsequently be added to increase the length of the resultant F1/R1 long product with primers F2/R2-LibAddition. Under the scheme depicted, the average length of library fragments below the cutoff and selected in the Small Library Fraction would reduce to $<\sim 200$ bp for F1/R1 products and to $<\sim 150$ bp for the F2/R2 products. Given that the average length of the cfDNA fragments preserved in original library were ~ 166 bp (see FIG. 33 ($\sim 229-63=166$ bp)), size selection of the F1/R1-LibAddition products would shift the cutoff and have the effect of reducing the average length of preserved cfDNA recovered in the Small Library Fraction to $<\sim 116$ bp. An even further reduction in length would be had for F2/R2 products, where average cfDNA length preserved would fall to $<\sim 66$ bp.

(184) While the ability to increase the representation of small cfDNA fragments in given sample at the time of purification by NSI-SSAP or once converted to an immortalized representative library by NSI-LSS has been tested directly and demonstrated herein using the increases in child fraction estimate as a readout, the method is fully translatable to conceivably any non-invasive test that queries cfDNA. These would include circulating tumor DNA, circulating pathogen DNA, circulating DNA from a transplanted organ. In addition, the methods would extent to circulating tumor mRNA, miRNA, lncRNA once converted to dsDNA.

(185) As used herein, the singular terms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to a molecule can include multiple molecules unless the context clearly dictates otherwise.

(186) As used herein, the terms “substantially,” “substantial,” and “about” are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, the terms can refer to less than or equal to $\pm 10\%$, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$.

(187) Additionally, amounts, ratios, and other numerical values are sometimes presented herein in a range format. It is to be understood that such range format is used for convenience and brevity and should be understood flexibly to include numerical values explicitly specified as limits of a range, but also to include all individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly specified. For example, a ratio in the range of about 1 to about 200 should be understood to include the explicitly recited limits of about 1 and about 200, but also to include individual ratios such as about 2, about 3, and about 4, and sub-ranges such as about 10 to about 50, about 20 to about 100, and so forth.

(188) In the foregoing description, it will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to the invention disclosed herein without departing from the scope and spirit of the invention. The invention illustratively described herein suitably

may be practiced in the absence of any element or elements, limitation or limitations, which is not specifically disclosed herein. The terms and expressions which have been employed are used as terms of description and not of limitation, and there is no intention that in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention. Thus, it should be understood that although the present invention has been illustrated by specific embodiments and optional features, modification and/or variation of the concepts herein disclosed may be resorted to by those skilled in the art, and that such modifications and variations are considered to be within the scopes of this invention.

Claims

1. A method for isolating nucleic acids from a biological sample, comprising: (a) contacting a first composition comprising nucleic acids in a biological sample with a first solid phase nucleic acid binding matrix to obtain a filtrate and a first eluate, wherein the first composition comprises: (i) less than 1% aliphatic alcohols, (ii) about 3.2055 M to about 4.76 M of a chaotropic compound, (iii) about 0% to about 12.1% of a nitrile compound, (iv) a pH of about 6.6 to about 7.157, (v) 0% to about 9% of a detergent, and (vi) 0% to about 1.2% of a chelating agent; wherein the first composition permits binding of greater than 30% of nucleic acids longer than about 194 bp to the first matrix, wherein the filtrate comprises remainder nucleic acids passed through the first matrix, and wherein the first eluate comprises nucleic acids bound and released from the first matrix; and (b) contacting a second composition comprising the filtrate comprising remainder nucleic acids from the first composition with a second solid phase nucleic acid binding matrix to obtain a second eluate, wherein the second composition comprises: (i) less than 1% aliphatic alcohols, (ii) about 3.2055 M to about 4.76 M of a chaotropic compound, (iii) about 20% to about 30% of a nitrile compound, (iv) a pH of about 4.05 to about 5.5, (v) 0% to about 9.55% of a detergent, and (vi) about 0 to about 0.61 mM of a chelating agent; wherein the second composition binds more than 40% of nucleic acids longer than about 72 bp to the second matrix, wherein the second eluate comprises nucleic acids bound and released from the second matrix; wherein the first and the second solid phase nucleic acid binding matrices comprise siliceous materials, silica gel, glass, glass fiber, or gelatinous silica.
2. The method of claim 1, further comprising washing the first matrix with a washing buffer after step (a), and eluting nucleic acids from the first matrix with an elution buffer to obtain the first eluate.
3. The method of claim 1, wherein the second eluate is obtained by washing the second matrix with a washing buffer after the contacting in step (b), and eluting nucleic acids from the second matrix with an elution buffer.
4. The method of claim 1, further comprising digesting the biological sample with a protease prior to step (a).
5. The method of claim 1, wherein the nitrile compound of the first and/or second composition comprises acetonitrile (ACN), propionitrile (PCN), butyronitrile (BCN), isobutylnitrile (IBCN), or a combination thereof; and wherein the chaotropic compound of the first and/or second composition comprises guanidine chloride (GnCl), urea, thiourea, guanidine thiocyanate, NaI, guanidine isothiocyanate, D-/L-arginine, a perchlorate or perchlorate salt of Li⁺, Na⁺, K⁺, or a combination thereof.
6. The method of claim 1, wherein the first composition comprises ACN as the nitrile compound, and about 3.2055 M to about 4 M of GnCl as the chaotropic compound; and wherein the second composition comprises ACN as the nitrile compound and about 3.5 M to about 4.76 M of GnCl as the chaotropic compound.
7. The method of claim 1, wherein the detergent comprises Triton X-100, Tween 20, N-lauroyl

sarcosine, sodium dodecylsulfate (SDS), dodecyldimethylphosphine oxide, sorbitan monopalmitate, decylhexaglycol, 4-nonylphenyl-polyethylene glycol, or a combination thereof; and wherein the chelating compound comprises ethylenediaminetetracetic (EDTA), ethyleneglycol-bis (2-aminoethylether)-N,N,N',N'-tetraacetic acid (EGTA), citric acid, N,N,N',N'-Tetrakis(2-pyridylmethyl)ethylenediamine (TPEN), 2,2'-Bipyridyl, deferoxamine methanesulfonate salt (DFOM), 2,3-Dihydroxybutanedioic acid (tartaric acid), or a combination thereof; and wherein the first composition and/or the second composition comprises less than 1% of alcohol.

8. The method of claim 1, wherein the biological sample is a plasma sample from a pregnant woman comprising fetal cfDNA and maternal cfDNA, or a plasma sample from a cancer patient comprising circulating tumor DNA; wherein when the biological sample is a plasma sample from a pregnant woman, fetal fraction of nucleic acids elutable from the second solid phase nucleic acid binding matrix is at least 1% higher than fetal fraction of nucleic acids elutable from the first solid phase nucleic acid binding matrix.

9. The method of claim 1, wherein step (a) binds more than 40% of nucleic acids longer than about 194 bp to the first solid phase nucleic acid binding matrix, and wherein step (b) binds more than 80% of nucleic acids longer than about 72 bp to the second solid phase nucleic acid binding matrix.

10. The method of claim 1, wherein the first composition further comprises Tris(hydroxymethyl) aminomethane (Tris-base), a chelating compound, and a detergent.

11. The method of claim 10, wherein the first composition comprises the biological sample in an amount from 38.9% to about 45.66%.

12. The method of claim 10, wherein the nitrile compound is acetonitrile, the chelating compound is EDTA, the detergent is Tween 20, and the chaotropic compound is guanidine chloride; wherein the biological sample is a plasma sample from a pregnant woman comprising fetal cell-free DNA (cfDNA) and maternal cfDNA, or a plasma sample from a cancer patient comprising circulating tumor DNA (ctDNA); and wherein: the method further comprising amplifying the cfDNA or ctDNA prior to step (a) by ligating the cfDNA or the ctDNA to a plurality of DNA adapter molecules, wherein the DNA adapter molecules comprises common forward and reverse primer binding sites, and then amplifying the ligated cfDNA or ctDNA by using forward and reverse primers complementary to the common primer binding sites in the DNA adaptor molecules, or the method further comprising adding a plurality of DNA base pairs to the cfDNA or ctDNA fragments by tailing PCR to increase the size of the cfDNA or ctDNA fragments prior to step (a).

13. The method of claim 1, wherein the second composition comprises 2-(N-morpholino) ethanesulfonic acid (MES).

14. The method of claim 13, wherein the second composition comprises, the MES in an amount from about 4.59 mM to about 38 mM.
